

Buy Low, Sell High: The Case for Equity QE*

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Abstract

Does the composition of central bank asset purchases affect profitability and welfare? We study this question in a New Keynesian model with two household types, Gertler-Karadi financial intermediaries with asset-specific divertability, long-term government bonds, equities, and an explicit central bank balance sheet with profit accounting. For demand and monetary policy shocks that drive the economy to the zero lower bound, equity-based quantitative easing generates substantially smaller losses than bond-based QE—and near-zero losses under active unwinding. The mechanism is an asset-price asymmetry: bond prices are elevated at the ZLB (the central bank buys high), while equity prices are depressed (it buys low). Welfare effects are small but systematically favor equity QE for restricted households via the fiscal channel. Our results provide a structural explanation for the divergent portfolio outcomes of the Bank of Japan’s equity purchases versus Western central banks’ bond-heavy programs.

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1 Introduction

Between 2022 and 2024, central banks across the developed world confronted an uncomfortable reckoning. As policy rates rose sharply to combat inflation, the bond portfolios accumulated during successive rounds of quantitative easing (QE) suffered dramatic mark-to-market losses. The Federal Reserve’s cumulative deferred asset—effectively its unrealised loss—exceeded \$200 billion. The Bank of England required an explicit Treasury indemnity to absorb over £100 billion in projected losses on its Asset Purchase Facility. The European Central Bank, the Reserve Bank of Australia, and the Bank of Canada all reported substantial negative equity positions. Meanwhile, the Bank of Japan—the only major central bank to have purchased equities on a large scale—held unrealised gains exceeding ¥20 trillion on its equity exchange-traded fund (ETF) portfolio, even as its bond holdings also faced valuation pressures. Was this divergence a coincidence, or does the *composition* of asset purchases systematically affect central bank profitability?

This paper asks a precise question: is there a systematic relationship between the composition of QE—long-term government bonds versus equities—and central bank profitability? If so, does this matter for household welfare through the fiscal channel that links central bank profits and losses to government revenues and taxes?

The intuition is straightforward and rests on an asset-price asymmetry. At the zero lower bound (ZLB), long-term bond prices are elevated because yields are compressed to near-zero levels. A central bank that purchases bonds during QE is therefore “buying high.” When the economy recovers and interest rates normalise, bond prices fall—the central bank is forced to “sell low” (or, if it holds to maturity, earns below-market coupons while paying rising interest on reserves). Equities, by contrast, are typically depressed during recessions that trigger the ZLB. A central bank purchasing equities at the trough is “buying low” and benefits from capital appreciation during the recovery. This buy-low-sell-high logic for equities stands in sharp contrast to the buy-high-sell-low dynamic for bonds.

The profitability asymmetry is not, however, universal. It depends critically on the nature of the shock driving the economy to the ZLB. Demand shocks (such as preference or discount factor shocks) and certain monetary policy shocks depress both output and equity valuations while compressing bond yields—producing the asymmetry described above. Supply shocks, by contrast, raise inflation while reducing output, which can push bond prices *down* during the recession itself, potentially reversing the profitability ranking. Understanding which shocks generate the asymmetry, and which do not, is essential for drawing policy conclusions.

We investigate these questions in a medium-scale New Keynesian DSGE model that combines several ingredients from the existing literature into a unified framework. The model

features two types of households—unrestricted households who participate directly in equity and bond markets, and restricted (hand-to-mouth) households who can only save via bank deposits—creating the market segmentation necessary to break Wallace neutrality and give QE real effects. Financial intermediaries in the tradition of [Gertler and Karadi \(2011, 2013\)](#) face leverage constraints with asset-specific divertability parameters, so that equities and bonds generate different credit spreads and respond differently to central bank purchases. The asset space includes short-term deposits, long-term government bonds modelled as Woodford perpetuities ([Woodford, 2001](#); [Chen et al., 2012](#)), and equity claims on the aggregate capital stock. Crucially, the central bank maintains an explicit balance sheet with period-by-period profit accounting, remittance rules, and a fiscal authority whose tax policy responds to central bank transfers.

Our main results can be organised around four findings. First, equity QE generates substantially smaller present-value losses than bond QE for shocks that drive the economy to the ZLB. Under a risk premium shock calibrated to produce a 6-quarter ZLB episode, the present value of central bank profits is -5.034 under equity QE with passive unwinding, compared to -9.008 under bond QE—a 44 percent reduction in losses. With active unwinding, the result is even more striking: *equity QE generates a net profit* of $+0.584$, while bond QE still loses -6.770 . Active unwinding transforms equity QE from a loss-making to a profit-making operation.

Second, the mechanism operates through the asset-price asymmetry described above. Equity prices (Tobin’s Q) fall during the recession, so the central bank acquires equity at depressed valuations. As the economy recovers, equity prices rise and the central bank realises capital gains upon unwinding. Bond prices, conversely, are elevated during the ZLB period (compressed yields) and fall as monetary policy normalises. The central bank’s bond portfolio generates negative capital gains net of interest costs. The profitability advantage of equity QE is especially pronounced under active unwinding, which crystallises the favourable equity price dynamics.

Third, QE itself deepens the recession through the fiscal channel. The output trough under no QE is -3.4 percent, but worsens to -5.0 percent with bond QE and -4.2 percent with equity QE, as QE losses feed back through higher distortionary labor taxes. This fiscal feedback effect is a novel finding: more profitable QE compositions produce a smaller contractionary fiscal drag. The distributional dimension is also systematic: restricted (hand-to-mouth) households bear disproportionate welfare costs from QE through the tax channel, and equity QE—by generating smaller losses—mitigates this burden.

Fourth, only the risk premium shock drives the economy to the ZLB under the estimated parameters. Six other shocks—including preference, monetary policy, financial, TFP,

government spending, and investment efficiency—do not trigger the ZLB at the calibrated magnitudes, so QE is never activated. This result highlights that the case for equity QE rests specifically on financial-crisis-type ZLB episodes, where the bond-equity price asymmetry is most pronounced.

This paper contributes to several strands of the literature. It is, to our knowledge, the first to study central bank profitability as a function of QE composition in a structural general equilibrium model with an explicit central bank balance sheet. The existing QE literature—including the seminal contributions of [Gertler and Karadi \(2011, 2013\)](#), [Sims and Wu \(2021\)](#), and [Chen et al. \(2012\)](#)—has focused on macroeconomic stabilisation (output, inflation) rather than the central bank’s own financial position. The central bank solvency literature ([Reis, 2015](#); [Hall and Reis, 2015](#); [Del Negro and Sims, 2015](#)) has analysed the fiscal consequences of central bank losses but in reduced-form settings that abstract from composition choices. Our model combines equities and bonds in a QE framework, allowing for a direct comparison that neither strand has previously undertaken.

Our household structure builds on the two-agent New Keynesian (TANK) framework of [Galí et al. \(2007\)](#) and its application to QE by [Cantore and Meichtry \(2024\)](#) and [Tsiaras \(2023\)](#). [Cantore and Meichtry \(2024\)](#) study QE unwinding with two household types but restrict the asset space to government bonds. [Tsiaras \(2023\)](#) incorporates both TANK households and Gertler-Kiyotaki financial intermediaries in a QE model but likewise excludes equities. [Hohberger et al. \(2020\)](#) include both equities and two household types but rely on portfolio-rebalancing frictions rather than leverage constraints, and do not track central bank profitability. We combine the multi-asset structure with intermediary leverage constraints, household heterogeneity, and explicit profit accounting—a combination that is new to the literature.

The paper also speaks to the empirical literature on the Bank of Japan’s equity ETF purchase program. [Hosono and Isobe \(2014\)](#) and [Barbon and Gianinazzi \(2019\)](#) document the market effects of the BoJ’s equity purchases, while [Farmer \(2012\)](#) provides an early policy proposal for central bank equity purchases, albeit without a formal general equilibrium analysis. Our model provides a structural explanation for why the BoJ’s equity portfolio generated large unrealised gains while Western central banks’ bond-heavy portfolios generated losses: the asset-price asymmetry is a feature of the macroeconomic environment at the ZLB, not an accident of market timing.

The remainder of the paper is organised as follows. Section 2 presents the model. Section 3 discusses calibration, Bayesian estimation, and steady-state properties. Section 4 presents the main results on profitability, macroeconomic stabilisation, and welfare. Section 5 conducts sensitivity analysis. Section 6 discusses policy implications, presents a

simple counterfactual exercise using actual Federal Reserve purchase data, and addresses limitations. Section 7 concludes.

2 Model

2.1 Overview

The economy is populated by four types of agents: households, firms, financial intermediaries, and the government (comprising a central bank and a fiscal authority). Time is discrete and indexed by $t = 0, 1, 2, \dots$

There are three financial assets. *Short-term deposits* are one-period claims issued by financial intermediaries, paying the real deposit rate R_t . *Long-term government bonds* are perpetuities with geometrically decaying coupons, following Woodford (2001) and Chen et al. (2012). *Equities* are claims on the aggregate capital stock, issued by capital goods producers.

Households are divided into two types. A fraction ψ_u are *unrestricted* households who can trade deposits, long bonds, and equities. The remaining fraction $1 - \psi_u$ are *restricted* (hand-to-mouth) households who consume their entire income each period and hold no financial assets. This market segmentation, together with the financial intermediary's leverage constraint, breaks Wallace neutrality and gives QE real effects.

The production sector has three layers: competitive final goods producers, monopolistically competitive intermediate goods firms with Rotemberg pricing, and competitive capital goods producers who convert final goods into installed capital subject to investment adjustment costs.

Financial intermediaries accept deposits from unrestricted households and invest in long bonds and equities. They face a moral hazard problem with asset-specific divertability parameters λ_{eq} and λ_L , generating endogenous leverage limits and credit spreads that differ across asset classes.

The central bank sets the short-term nominal rate via a Taylor rule subject to a zero lower bound, conducts asset purchases when the ZLB binds, and maintains an explicit balance sheet with profit/loss accounting. The fiscal authority levies a distortionary labor income tax whose rate responds to government debt and central bank remittances, following Schmitt-Grohé and Uribe (2004) and Leeper et al. (2010).

2.2 Households

2.2.1 Unrestricted Households

A fraction ψ_u of households have unrestricted access to all financial markets. They choose consumption $C_{u,t}$, labour supply $L_{u,t}$, equity holdings $S_{u,t}$, and long bond holdings $B_{u,t}^L$ to maximise

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\mathcal{A}_t \frac{C_{u,t}^{1-\sigma_c}}{1-\sigma_c} - \zeta_U \frac{L_{u,t}^{1+\varphi}}{1+\varphi} \right] \quad (1)$$

where β is the discount factor, σ_c is the coefficient of relative risk aversion, φ is the inverse Frisch elasticity of labour supply, and ζ_U is a labour disutility weight calibrated to match a steady-state labour target.

The preference shock $\mathcal{A}_t$ enters *only* the consumption term—not the labour disutility—so that utility is separable in consumption and leisure. This specification ensures that a negative preference shock reduces both the marginal utility of consumption and the shadow value of income from working, generating a proper demand shortfall that can push the economy to the ZLB.¹

The budget constraint is

$$C_{u,t} + Q_t^K (1 + \gamma_S) S_{u,t} + Q_t^L (1 + \gamma_B) B_{u,t}^L = W_t (1 - \tau_t^L) L_{u,t} + (Z_t^K + (1 - \delta) Q_t^K) S_{u,t-1} + \frac{1 + \chi Q_t^L}{\Pi_t} B_{u,t-1}^L + \Pi_{f,t} \quad (2)$$

where Q_t^K is the equity price (Tobin's Q), Q_t^L is the long bond price, W_t is the real wage, τ_t^L is the labor income tax rate, Z_t^K is the capital rental rate, δ is the depreciation rate, χ is the coupon decay parameter, and Π_t is gross inflation. Deposits are eliminated via Walras' Law.

The holding cost parameters γ_S and γ_B create permanent wedges between asset returns and the deposit rate. They serve two purposes: generating realistic steady-state credit spreads and preventing unrestricted households from perfectly offsetting central bank purchases (thereby preserving QE effectiveness even among market participants).²

The first-order conditions yield the marginal utility of consumption,

$$U_{c,u,t} = \mathcal{A}_t \cdot C_{u,t}^{-\sigma_c}, \quad (3)$$

¹With non-separable utility where $\mathcal{A}_t$ multiplies both terms, the preference shock cancels from the intratemporal labour-leisure condition and has negligible output effects.

²These proportional holding costs are a simplified version of the quadratic portfolio adjustment costs discussed in [Gertler and Karadi \(2013\)](#). Both formulations break Wallace neutrality and generate spreads; the proportional form yields a cleaner analytical steady state.

the labour supply condition,

$$W_t(1 - \tau_t^L) \cdot U_{c,u,t} = \zeta_U \cdot L_{u,t}^\varphi, \quad (4)$$

the deposit Euler equation,

$$U_{c,u,t} = \beta \cdot R_t \cdot \mathbb{E}_t [U_{c,u,t+1}], \quad (5)$$

the equity Euler equation,

$$U_{c,u,t} \cdot Q_t^K (1 + \gamma_S) \cdot v_{rp,t} = \beta \cdot \mathbb{E}_t [U_{c,u,t+1} (Z_{t+1}^K + (1 - \delta)Q_{t+1}^K)], \quad (6)$$

where $v_{rp,t}$ is a time-varying risk premium wedge ($v_{rp,t} = 1$ in steady state), and the long bond Euler equation,

$$U_{c,u,t} \cdot Q_t^L (1 + \gamma_B) = \beta \cdot \mathbb{E}_t \left[U_{c,u,t+1} \cdot \frac{1 + \chi Q_{t+1}^L}{\Pi_{t+1}} \right]. \quad (7)$$

The stochastic discount factor of the unrestricted household is

$$\Lambda_{u,t} \equiv \beta \cdot \frac{U_{c,u,t+1}}{U_{c,u,t}}. \quad (8)$$

2.2.2 Restricted Households

A fraction $1 - \psi_u$ of households are hand-to-mouth. They choose consumption $C_{r,t}$ and labour $L_{r,t}$ to maximise

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\mathcal{A}_t \frac{C_{r,t}^{1-\sigma_c}}{1-\sigma_c} - \zeta_R \frac{L_{r,t}^{1+\varphi}}{1+\varphi} \right] \quad (9)$$

subject to the period budget constraint

$$C_{r,t} = W_t(1 - \tau_t^L)L_{r,t} + \Pi_{f,t}. \quad (10)$$

Restricted households have no access to equity, bond, or deposit markets; they consume their entire period income. This is the strongest form of market non-participation, standard in TANK models following Galí et al. (2007). The marginal utility and labour supply conditions mirror equations (3) and (4) with ζ_R replacing ζ_U .

Restricted households play two roles. First, they break Wallace neutrality: being unable to rebalance portfolios in response to central bank purchases, they ensure that QE creates genuine asset scarcity. Second, they amplify the fiscal channel: as hand-to-mouth agents, their after-tax labour income responds directly to changes in τ_t^L , making the welfare

consequences of central bank profitability or losses particularly direct. When central bank losses reduce remittances and the fiscal rule raises τ_t^L , restricted households bear the full consumption impact through the after-tax wage wedge $W_t(1 - \tau_t^L)$.

2.2.3 Aggregation

Aggregate consumption and labour are population-weighted averages:

$$C_t = \psi_u C_{u,t} + (1 - \psi_u) C_{r,t}, \quad L_{agg,t} = \psi_u L_{u,t} + (1 - \psi_u) L_{r,t}. \quad (11)$$

2.3 Production

2.3.1 Final and Intermediate Goods

A competitive final goods producer aggregates a continuum of intermediate varieties using the CES technology $Y_t = \left[\int_0^1 Y_{i,t}^{(\epsilon-1)/\epsilon} di \right]^{\epsilon/(\epsilon-1)}$, where $\epsilon > 1$ is the elasticity of substitution. The demand for each variety is $Y_{i,t} = (P_{i,t}/P_t)^{-\epsilon} Y_t$. The steady-state markup is $\epsilon/(\epsilon - 1)$.

Each intermediate firm i produces using a Cobb-Douglas technology:

$$Y_t = Z_{tfp,t} \cdot K_{t-1}^\alpha \cdot L_{agg,t}^{1-\alpha} \quad (12)$$

where α is the capital share and $Z_{tfp,t}$ is aggregate TFP. Cost minimisation yields the factor demand conditions

$$Z_t^K = \alpha \cdot MC_t \cdot \frac{Y_t}{K_{t-1}}, \quad W_t = (1 - \alpha) \cdot MC_t \cdot \frac{Y_t}{L_{agg,t}}, \quad (13)$$

where MC_t is real marginal cost.

Intermediate firms face Rotemberg (1982) quadratic price adjustment costs. In a symmetric equilibrium, the New Keynesian Phillips Curve (NKPC) is

$$\psi_p \Pi_t (\Pi_t - 1) = (1 + \tau_S)(1 - \epsilon) + \epsilon \cdot MC_t + \beta \frac{U_{c,u,t+1}}{U_{c,u,t}} \psi_p \Pi_{t+1} (\Pi_{t+1} - 1) \frac{Y_{t+1}}{Y_t} \quad (14)$$

where ψ_p is the Rotemberg cost parameter (calibrated to match the Calvo equivalent), $\tau_S = 1/(\epsilon - 1)$ is a production subsidy that eliminates the steady-state markup distortion, and the unrestricted household's SDF prices the firm (as firms are owned by unrestricted households). Distributed profits are

$$\Pi_{f,t} = \left(1 - MC_t - \frac{\psi_p}{2} (\Pi_t - 1)^2 \right) Y_t. \quad (15)$$

2.3.2 Capital Goods Producers and Equity

Capital goods producers own the physical capital stock and issue equity claims. They convert final goods into new capital subject to investment adjustment costs, with the transformation scaled by an investment efficiency shock $\Upsilon_{inv,t}$ ($\Upsilon_{inv,t} = 1$ in steady state):

$$K_t = \Upsilon_{inv,t} \left[1 - \frac{\zeta_{inv}}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \right] I_t + (1 - \delta)K_{t-1} \quad (16)$$

where ζ_{inv} governs the curvature of adjustment costs. The capital goods producer's first-order condition for investment yields the Tobin's Q equation:

$$Q_t^K \Upsilon_{inv,t} \left[1 - \frac{\zeta_{inv}}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 - \zeta_{inv} \left(\frac{I_t}{I_{t-1}} - 1 \right) \frac{I_t}{I_{t-1}} \right] + \mathbb{E}_t \left[\Lambda_{u,t} Q_{t+1}^K \Upsilon_{inv,t+1} \zeta_{inv} \left(\frac{I_{t+1}}{I_t} - 1 \right) \left(\frac{I_{t+1}}{I_t} \right)^2 \right] = 1 \quad (17)$$

At steady state, $I_t = I_{t-1}$ implies $Q_{ss}^K = 1$: the price of installed capital equals the price of new investment goods.

The gross equity return—the total holding-period return on one unit of capital—is

$$R_t^{eq} = \frac{Z_t^K + (1 - \delta)Q_t^K}{Q_{t-1}^K}. \quad (18)$$

Total equity outstanding equals the capital stock: one share per unit of installed capital. Shares are held by unrestricted households ($\psi_u S_{u,t}$ in aggregate), financial intermediaries ($S_{fi,t}$), and—during QE episodes—the central bank (S_t^{CB}).

2.4 Long-Term Government Bonds

Long-term bonds are modelled as Woodford perpetuities with geometrically decaying coupons. A bond purchased at time t pays χ^{j-1} units of the numeraire at time $t + j$, for $j = 1, 2, \dots$, where $\chi \in [0, 1)$ controls duration. The key property is that a bond purchased at t that has made its coupon at $t + 1$ is identical to χ units of a newly issued bond, so a single price Q_t^L summarises the entire portfolio.

The gross nominal return on long bonds is

$$R_t^{Ln} = \frac{1 + \chi Q_t^L}{Q_{t-1}^L}, \quad (19)$$

and the gross real return is $R_t^L = R_t^{Ln} / \Pi_t$. The Fisher equation relates the nominal and real policy rates: $R_t = R_t^n / \Pi_{t+1}$.

The Macaulay duration of the perpetuity is $D = R_{ss}^{Ln}/(R_{ss}^{Ln} - \chi)$. With $\chi = 0.95$ and the calibrated parameters, this yields a duration of approximately 15 quarters (roughly 3.8 years).

The credit spreads on equities and long bonds are defined as

$$\text{spread}_t^{eq} = \mathbb{E}_t[R_{t+1}^{eq}] - R_t, \quad \text{spread}_t^L = \mathbb{E}_t[R_{t+1}^L] - R_t. \quad (20)$$

In steady state, the holding costs generate spreads $\text{spread}_{ss}^{eq} = \gamma_S/\beta$ and $\text{spread}_{ss}^L = \gamma_B/\beta$, consistent with the intermediary's asset-specific divertability (see Section 2.5).

2.5 Financial Intermediaries

Financial intermediaries follow the [Gertler and Karadi \(2011, 2013\)](#) framework, extended to allow for asset-specific divertability. A continuum of bankers operate intermediaries. Each period, a fraction $1 - \sigma_b$ of bankers exit and transfer their net worth to unrestricted households; they are replaced by new bankers who receive a startup transfer ω_{fi} times total intermediary assets.

2.5.1 Balance Sheet and Net Worth

The intermediary's balance sheet is

$$Q_t^K S_{fi,t} + Q_t^L B_{fi,t}^L = N_{fi,t} + D_t, \quad (21)$$

where $S_{fi,t}$ denotes equity holdings, $B_{fi,t}^L$ long bond holdings, $N_{fi,t}$ net worth, and D_t deposits. Net worth evolves according to

$$N_{fi,t} = \xi_{fin,t} [(\sigma_b + \omega_{fi}) (R_t^{eq} Q_{t-1}^K S_{fi,t-1} + R_t^L Q_{t-1}^L B_{fi,t-1}^L) - \sigma_b R_{t-1} D_{t-1}] \quad (22)$$

where $\xi_{fin,t}$ is a financial shock that scales the effective return on intermediary assets.

2.5.2 Incentive Constraint and Leverage

Bankers can divert a fraction of their assets and default on depositors. Crucially, the divertable fraction is *asset-specific*: a fraction λ_{eq} of equity holdings and a fraction λ_L of bond holdings can be diverted, where $\lambda_{eq} > \lambda_L$ reflects the greater information asymmetry and lower liquidity of equities. Depositors will fund the banker only if the franchise value exceeds the temptation to divert:

$$V_t \geq \lambda_{eq} Q_t^K S_{fi,t} + \lambda_L Q_t^L B_{fi,t}^L. \quad (23)$$

Following [Gertler and Karadi \(2011\)](#), the value function is conjectured to be linear in the banker's portfolio: $V_t = \nu_t^S Q_t^K S_{fi,t} + \nu_t^B Q_t^L B_{fi,t}^L + \eta_t N_{fi,t}$, where ν_t^S and ν_t^B are the marginal values of equity and bond holdings and η_t is the marginal value of net worth. These coefficients satisfy

$$\nu_t^S = \mathbb{E}_t[\beta \Lambda_{u,t+1} \Omega_{fi,t+1}] \cdot (R_{t+1}^{eq} - R_t), \quad \nu_t^B = \mathbb{E}_t[\beta \Lambda_{u,t+1} \Omega_{fi,t+1}] \cdot (R_{t+1}^L - R_t), \quad \eta_t = \mathbb{E}_t[\beta \Lambda_{u,t+1} \Omega_{fi,t+1}] \cdot R_t, \quad (24)$$

where $\Omega_{fi,t}$ is the shadow value of net worth, evolving recursively as

$$\Omega_{fi,t} = (1 - \sigma_b) + \sigma_b \cdot \mathbb{E}_t \left[\Lambda_{u,t+1} \Omega_{fi,t+1} \left(\text{spread}_t^{eq} \phi_{fi,t} x_{fi} + \text{spread}_t^L \phi_{fi,t} (1 - x_{fi}) + R_t \right) \right]. \quad (25)$$

The intermediary's portfolio composition x_{fi} (equity share of total assets) is fixed at its steady-state value, following [Gertler and Karadi \(2013\)](#) and [Sims and Wu \(2021\)](#). This resolves the portfolio indeterminacy that arises from the linear incentive constraint.³ The leverage ratio $\phi_{fi,t} = (Q_t^K S_{fi,t} + Q_t^L B_{fi,t}^L) / N_{fi,t}$ is determined by the binding incentive constraint:

$$\Lambda_{u,t+1} \Omega_{fi,t+1} R_t = (\lambda_{eq} - \Lambda_{u,t+1} \Omega_{fi,t+1} (R_{t+1}^{eq} - R_t)) \phi_{fi,t} x_{fi} + (\lambda_L - \Lambda_{u,t+1} \Omega_{fi,t+1} (R_{t+1}^L - R_t)) \phi_{fi,t} (1 - x_{fi}). \quad (26)$$

Deposits are determined residually: $D_t = (\phi_{fi,t} - 1) N_{fi,t}$.

2.5.3 QE Transmission Through the Intermediary Channel

Central bank asset purchases operate through two channels in this framework. First, by purchasing assets directly, the central bank removes them from the private sector, relaxing the intermediary leverage constraint and reducing credit spreads. Second, because $\lambda_{eq} \neq \lambda_L$, purchasing equities versus bonds has *different* effects on the effective leverage constraint per dollar spent. With $\lambda_{eq} > \lambda_L$, each dollar of equity purchased relaxes the constraint by more than a dollar of bonds, giving the QE composition real macroeconomic effects beyond the profitability channel.

³The economic interpretation is that intermediaries maintain a fixed asset allocation reflecting regulatory requirements or slow-moving portfolio policies. The steady-state equity share $x_{fi,ss}$ is determined endogenously by market clearing.

2.6 Central Bank

2.6.1 Monetary Policy Rule

The central bank sets the nominal interest rate according to an inertial Taylor rule subject to a zero lower bound:

$$R_t^n = \max \{1, R_t^{n,unc}\} \quad (27)$$

where the unconstrained rate follows

$$\ln \left(\frac{R_t^{n,unc}}{R_{ss}^n} \right) = \rho_r \ln \left(\frac{R_{t-1}^n}{R_{ss}^n} \right) + (1 - \rho_r) \left[\phi_\pi \ln \left(\frac{\Pi_t}{\Pi_{ss}} \right) + \phi_y \ln \left(\frac{Y_t}{Y_{ss}} \right) \right] + \varepsilon_t^M \quad (28)$$

with ρ_r governing interest rate smoothing, ϕ_π the inflation response, ϕ_y the output gap response, and ε_t^M a monetary policy shock. The ZLB constraint is implemented using the Oc-cBin methodology of [Guerrieri and Iacoviello \(2015\)](#), which solves the model as a piecewise-linear system alternating between constrained ($R_t^n = 1$) and unconstrained ($R_t^n = R_t^{n,unc}$) regimes. The solver endogenously determines the timing and duration of ZLB episodes.

2.6.2 Asset Purchase Rules

When the ZLB binds, the central bank purchases long bonds and/or equities. The purchase rules are

$$B_t^{CB} = (\chi - d_t^{unwind} \rho_B^{unwind}) B_{t-1}^{CB} + d_t^{purchase} \Phi_B B_t^L, \quad (29)$$

$$S_t^{CB} = (1 - d_t^{unwind} \rho_E^{unwind}) S_{t-1}^{CB} + d_t^{purchase} \Phi_E K_{t-1}, \quad (30)$$

where $d_t^{purchase}$ is an indicator equal to one during the ZLB episode, d_t^{unwind} is an indicator equal to one during the active unwinding phase, and Φ_B and Φ_E control the aggressiveness of bond and equity purchases as a fraction of outstanding supply. The four QE regimes are: (i) no QE ($\Phi_B = \Phi_E = 0$); (ii) bond-only QE ($\Phi_B > 0, \Phi_E = 0$); (iii) equity-only QE ($\Phi_B = 0, \Phi_E > 0$); and (iv) mixed QE ($\Phi_B, \Phi_E > 0$).

2.6.3 Unwinding Strategies

After the ZLB episode ends, two unwinding strategies are considered.

Passive (hold to maturity): Bonds run off naturally at rate $1 - \chi$ per period; equities are held indefinitely. No active sales occur. This minimises realised capital losses but keeps the balance sheet large for an extended period.

Active (gradual sell-off): After interest rate liftoff, the central bank sells a fraction ρ_B^{unwind}

of bonds and ρ_E^{unwind} of equities each period:

$$B_t^{CB} = (1 - \rho_B^{unwind})B_{t-1}^{CB}, \quad S_t^{CB} = (1 - \rho_E^{unwind})S_{t-1}^{CB}. \quad (31)$$

Active unwinding crystallises capital gains on equities (which have appreciated during the recovery) and capital losses on bonds (whose prices have fallen as rates normalise).

2.6.4 Profit Accounting

The central bank's period- t operating profit is

$$\text{ProfCB}_t = \underbrace{R_t^{eq} Q_{t-1}^K S_{t-1}^{CB}}_{\text{Equity return}} + \underbrace{R_t^L Q_{t-1}^L B_{t-1}^{CB}}_{\text{Bond return}} - \underbrace{R_{t-1} \cdot \text{Res}_{t-1}}_{\text{Reserve interest}} - \underbrace{Q_t^K (S_t^{CB} - S_{t-1}^{CB})}_{\text{Net equity purchases}} - \underbrace{Q_t^L (B_t^{CB} - B_{t-1}^{CB})}_{\text{Net bond purchases}}. \quad (32)$$

Profits can be decomposed into a net interest income component (dividends and coupons less reserve interest) and a capital gains component (changes in the market value of holdings). The central bank remits all profits to the fiscal authority:

$$\text{Rem}_t = \text{ProfCB}_t. \quad (33)$$

This simple remittance rule creates a direct fiscal channel: when the central bank earns profits, remittances reduce the tax burden; when it incurs losses, taxes must rise to compensate.

The present value of central bank profits—our primary profitability metric—is

$$PV_t^{CB} = \sum_{j=0}^T R_{ss}^{-j} \cdot \text{ProfCB}_{t+j}, \quad (34)$$

computed over the simulation horizon T , discounted at the steady-state real rate.

2.7 Fiscal Authority

The government levies a distortionary labor income tax at rate τ_t^L , spends G_t on final goods, issues long-term bonds, and receives remittances from the central bank. The government budget constraint (dropped from the model by Walras' Law, but verified to hold) is

$$G_t + \frac{1 + \chi Q_t^L}{\Pi_t} B_{t-1}^L = \tau_t^L W_t L_{agg,t} + Q_t^L B_t^L + \text{Rem}_t. \quad (35)$$

The labor tax rate follows a fiscal rule with smoothing, a debt-stabilising feedback, and

a remittance passthrough term, in the spirit of [Schmitt-Grohé and Uribe \(2004\)](#) and [Leeper et al. \(2010\)](#):

$$\tau_t^L - \tau_{ss}^L = \rho_\tau (\tau_{t-1}^L - \tau_{ss}^L) + (1 - \rho_\tau) \phi_\tau \left(\frac{Q_{t-1}^L B_{t-1}^L}{Y_{ss}} - \frac{Q_{ss}^L B_{ss}^L}{Y_{ss}} \right) - \phi_{Rem} \frac{Rem_t}{Y_{ss}}, \quad (36)$$

where ρ_τ governs tax smoothing, ϕ_τ controls the response to market-value government debt deviations (ensuring fiscal solvency), and ϕ_{Rem} determines the passthrough of central bank remittances to the tax rate. The debt feedback operates on the market value of outstanding bonds $Q_t^L B_t^L / Y_{ss}$ rather than face value, so that bond price movements—driven in part by monetary policy—feed back to fiscal policy. Government spending follows an AR(1) process $\ln(G_t / G_{ss}) = \rho_G \ln(G_{t-1} / G_{ss}) - \varepsilon_t^G$, and long bond supply follows an AR(1) process $\ln(B_t^L / B_{ss}^L) = \rho_{BL} \ln(B_{t-1}^L / B_{ss}^L) + \varepsilon_t^{BL}$.

The distortionary nature of the labor tax is the linchpin connecting central bank profitability to real outcomes. The causal chain is: central bank losses $\rightarrow$ lower remittances $\rightarrow$ fiscal rule raises $\tau_t^L \rightarrow$ after-tax wage $W_t(1 - \tau_t^L)$ falls $\rightarrow$ labor supply declines $\rightarrow$ output falls. This channel operates through the intratemporal labor supply margin (equation 4), creating a genuine efficiency cost of central bank losses that is absent under lump-sum taxation. Equity QE, by generating smaller losses, induces a smaller tax distortion and therefore a shallower output trough—giving QE composition real macroeconomic effects beyond the balance sheet.

2.8 Market Clearing and Equilibrium

The resource constraint is

$$Y_t = C_t + I_t + G_t + \frac{\psi_p}{2} (\Pi_t - 1)^2 Y_t \quad (37)$$

where the last term captures the real resource cost of price adjustment.

Equity market clearing requires that total shares equal the capital stock:

$$\psi_u S_{u,t} + S_{fi,t} + S_t^{CB} = K_{t-1}. \quad (38)$$

Long bond market clearing requires

$$\psi_u B_{u,t}^L + B_{fi,t}^L + B_t^{CB} = B_t^L. \quad (39)$$

An equilibrium is a sequence of prices $\{W_t, Q_t^K, Q_t^L, R_t, R_t^n, \Pi_t\}$ and quantities $\{C_{u,t}, C_{r,t}, L_{u,t}, L_{r,t}, S_{u,t}, B_{u,t}, B_{r,t}, K_t\}$.

such that: (i) households optimise given prices; (ii) firms optimise given factor prices and demand; (iii) intermediaries optimise subject to the leverage constraint; (iv) the central bank follows its Taylor rule, QE rules, and profit accounting; (v) the fiscal authority follows its tax and spending rules; and (vi) all markets clear.

2.8.1 Shock Processes

The model includes seven structural shock processes. First, the four shocks from the core model:

$$\ln \mathcal{A}_t = \rho_A \ln \mathcal{A}_{t-1} - \varepsilon_t^A, \quad (40)$$

$$\ln Z_{tfp,t} = \rho_Z \ln Z_{tfp,t-1} + \varepsilon_t^Z, \quad (41)$$

$$\ln \xi_{fin,t} = \rho_{fin} \ln \xi_{fin,t-1} - \varepsilon_t^{Fin}, \quad (42)$$

$$\varepsilon_t^M \sim N(0, \sigma_M^2). \quad (43)$$

The preference shock ε_t^A reduces the marginal utility of consumption, generating a demand-driven ZLB episode. The TFP shock ε_t^Z is a standard supply disturbance. The financial shock ε_t^{Fin} scales intermediary net worth. The monetary policy shock ε_t^M is an i.i.d. innovation to the Taylor rule.

Three additional shocks enrich the model's dynamics and are important for estimation:

$$\ln(G_t/G_{ss}) = \rho_G \ln(G_{t-1}/G_{ss}) - \varepsilon_t^G, \quad (44)$$

$$\ln v_{rp,t} = \rho_{rp} \ln v_{rp,t-1} + \varepsilon_t^{RP}, \quad (45)$$

$$\ln \Upsilon_{inv,t} = \rho_{inv} \ln \Upsilon_{inv,t-1} - \varepsilon_t^{Inv}. \quad (46)$$

The government spending shock ε_t^G is a fiscal demand disturbance (a positive innovation reduces G_t). The risk premium shock ε_t^{RP} enters the unrestricted household's equity Euler equation (6) as a wedge between the deposit rate and equity returns; a positive innovation raises the required equity premium and depresses investment demand. The investment efficiency shock ε_t^{Inv} scales the transformation of investment goods into installed capital in equations (16) and (17); a positive innovation reduces the efficiency of investment, acting as a supply-side disruption to capital formation. Finally, long bond supply follows the AR(1) process $\ln(B_t^L/B_{ss}^L) = \rho_{BL} \ln(B_{t-1}^L/B_{ss}^L) + \varepsilon_t^{BL}$.

Sign conventions are chosen so that positive innovations to ε_t^A , ε_t^G , ε_t^{Fin} , and ε_t^{Inv} are contractionary, while positive innovations to ε_t^Z are expansionary. All AR(1) persistence parameters satisfy $\rho_j \in (0, 1)$.

3 Calibration

The model is calibrated to the US economy at quarterly frequency. We group parameters into standard macroeconomic parameters, monetary policy parameters, financial intermediary parameters, market segmentation parameters, and fiscal parameters. Table 1 summarises the baseline calibration.

3.1 Parameter Values

Preferences. The discount factor $\beta = 0.995$ implies a steady-state annualised real rate of approximately 2 percent. The inverse Frisch elasticity $\varphi = 2.0$ is standard. The coefficient of relative risk aversion σ_c is estimated (see Section 3.4); the posterior mode is 2.71, somewhat higher than the commonly calibrated value of 1.5 and consistent with the micro-econometric evidence surveyed in Havránek et al. (2015). The labour disutility weights ζ_U and ζ_R are calibrated endogenously to match a steady-state labour supply of unity for both household types.

Production. The capital share $\alpha = 0.33$, depreciation rate $\delta = 0.025$ (10 percent annual), and elasticity of substitution $\epsilon = 11$ (implying a 10 percent steady-state markup) are standard. A production subsidy $\tau_S = 1/(\epsilon - 1) = 0.10$ eliminates the markup distortion in steady state. Price stickiness follows a Rotemberg specification with cost parameter $\psi_p \approx 116.5$, equivalent to a Calvo probability of $\theta = 0.75$ (average price duration of four quarters). The investment adjustment cost parameter $\zeta_{inv} = 5.0$ follows Christiano et al. (2005).

Monetary policy. The output gap response $\phi_y = 0.125$ and steady-state gross inflation $\Pi_{ss} = 1.005$ (approximately 2 percent annualised) are calibrated. The inflation response ϕ_π and interest rate smoothing ρ_r are estimated; the posterior modes are $\phi_\pi = 2.62$ and $\rho_r = 0.91$, consistent with the strong inflation response and high degree of interest rate inertia found in post-Volcker Taylor rule estimates (Clarida et al., 2000).

Financial intermediaries. The banker survival rate $\sigma_b = 0.972$ implies an expected banker lifespan of $1/(1 - \sigma_b) \approx 36$ quarters (9 years). The equity divertability parameter $\lambda_{eq} = 0.50$ and bond divertability $\lambda_L = 0.381$ follow Gertler and Karadi (2013), who set $\lambda_{eq}/\lambda_L > 1$ to reflect the greater information asymmetry and lower liquidity of equities relative to government bonds. This differential is the key friction that makes QE composition matter for intermediary leverage. The new banker transfer ω_{fi} is calibrated endogenously to ensure consistency of the net worth evolution in steady state.

Market segmentation. The fraction of unrestricted households $\psi_u = 0.50$ is consistent with US stock market participation rates of approximately 58 percent (Survey of Consumer Finances, 2022). The equity holding cost $\gamma_S = 0.003$ generates an annualised equity spread

of approximately 121 basis points ($4 \times \gamma_S/\beta$). The bond holding cost is set to $\gamma_B = \gamma_S \cdot \lambda_L/\lambda_{eq} \approx 0.0023$, ensuring that the ratio of household-side spreads matches the intermediary-side divertability ratio. This generates an annualised bond spread of approximately 92 basis points.

Long bonds. The coupon decay rate $\chi = 0.95$ yields a Macaulay duration of approximately 15 quarters (3.8 years). While shorter than the typical maturity of QE-targeted bonds (7–10 years), this parametrisation balances computational tractability with meaningful interest rate sensitivity. The sensitivity analysis in Section 5 explores higher values of χ .

Fiscal. Government spending is 20 percent of GDP ($G/Y = 0.20$). Government debt is 60 percent of annual GDP ($B^L/4Y = 0.60$, or $B^L/Y = 2.40$ in quarterly terms). The steady-state labor tax rate $\tau_{ss}^L \approx 0.254$ is not a free parameter but is determined endogenously in the steady state to satisfy the government budget constraint, given G/Y and B^L/Y . This value is broadly consistent with US effective labor tax rates (Mendoza et al., 1994). The fiscal rule features smoothing ($\rho_\tau = 0.70$), a debt feedback coefficient ($\phi_\tau = 0.02$), and a remittance passthrough ($\phi_{Rem} = 0.10$). The debt feedback operates on market-value debt deviations, ensuring fiscal solvency, while the remittance passthrough creates the direct channel from central bank profitability to labor tax distortions.

QE parameters. The bond purchase intensity $\Phi_B = 0.001$ generates a peak balance sheet of approximately 20 percent of quarterly GDP under bond-only QE. The equity purchase intensity $\Phi_E = 0.004$ generates a comparable balance sheet (22 percent of quarterly GDP). For mixed QE, $\Phi_B = 0.0008$ and $\Phi_E = 0.002$ yield a combined peak balance sheet of approximately 27 percent of quarterly GDP. The active unwinding rates are $\rho_B^{unwind} = 0.05$ and $\rho_E^{unwind} = 0.02$ per quarter.

3.2 Steady-State Properties

Table 2 reports the model’s steady-state values for key variables. The output decomposition features a consumption share $C/Y = 0.58$, investment share $I/Y = 0.22$, and government spending share $G/Y = 0.20$, consistent with US national accounts data.

The annualised nominal interest rate in steady state is 6.2 percent ($R_{ss}^n = \Pi_{ss}/\beta$), the real rate 4.1 percent ($R_{ss} = 1/\beta$), the equity return 5.4 percent, and the long bond return 5.1 percent. The equity spread of 121 basis points and bond spread of 92 basis points imply a spread ratio of $\gamma_S/\gamma_B = \lambda_{eq}/\lambda_L \approx 1.31$, consistent by construction with the intermediary divertability parameters.

The financial intermediary sector features a leverage ratio $\phi_{fi} = 3.57$, meaning intermediaries hold assets equal to 3.57 times their net worth. The equity share of intermediary

Table 1: Baseline Calibration

Parameter	Description	Symbol	Value
<i>Preferences</i>			
Discount factor	CRRA	β	0.995
Risk aversion		σ_c	2.7
Inverse Frisch elasticity		φ	2.0
<i>Production</i>			
Capital share	quarterly	α	0.33
Depreciation rate		δ	0.025
Elasticity of substitution		ϵ	11
Price adjustment cost	Rotemberg	ψ_p	118.2
Investment adj. cost		ζ_{inv}	5.0
<i>Monetary Policy</i>			
Inflation response	Taylor rule	ϕ_π	2.58
Output response	Taylor rule	ϕ_y	0.125
Interest rate smoothing	quarterly	ρ_r	0.9
Steady-state inflation		Π_{ss}	1.0050
<i>Financial Intermediaries</i>			
Equity divertability	Woodford bonds	λ_{eq}	0.500
Bond divertability		λ_L	0.381
Banker survival rate		σ_b	0.972
Coupon decay rate		χ	0.95
<i>Market Segmentation</i>			
Unrestricted HH share	CB share sensitivity	ψ_u	0.50
Equity holding cost		γ_S	0.0030
Bond holding cost		γ_B	0.0023
Equity portfolio balance		κ_S	5.0
Bond portfolio balance		κ_B	5.0
<i>Fiscal</i>			
Government spending/GDP	calibrated	G/Y	0.20
Debt/annual GDP		$B/4Y$	0.60
SS labor tax rate		τ_{ss}^L	0.340
Tax smoothing	fiscal rule	ρ_τ	0.70
Debt response		ϕ_τ	0.020
Remittance passthrough		ϕ_{Rem}	0.10
<i>Shock Processes</i>			
TFP persistence		ρ_Z	0.90
Preference persistence		ρ_A	1.00
Financial persistence		ρ_{fin}	0.90
Gov. spending persistence		ρ_G	0.86
Risk premium persistence		ρ_{rp}	0.85
Inv. efficiency persistence		ρ_{inv}	0.25

assets is $x_{fi} = 0.196$ (approximately 20 percent), with the remainder in long bonds. Unrestricted households hold 30 percent of total equities and 30 percent of long bonds, with intermediaries holding the remaining 70 percent of each.

The long bond price $Q_{ss}^L \approx 14.82$ reflects the present value of the decaying coupon stream discounted at the long-run nominal bond return. The capital stock $K_{ss} \approx 25.07$ and Tobin's $Q_{ss}^K = 1$ at steady state.

In the steady state, the central bank holds no bonds or equities ($B^{CB} = S^{CB} = 0$), central bank profits and remittances are zero, and the dividend per unit of capital is $\text{Div}_{ss} = Z_{k,ss} \approx 0.013$.

Table 2: Steady-State Values

Variable	Symbol	Value
<i>Real Aggregates</i>		
Output	Y	3.1070
Consumption	C	1.7050
Investment	I	0.7760
Capital	K	31.0401
Consumption/GDP	C/Y	0.55
Investment/GDP	I/Y	0.25
<i>Interest Rates (annualized)</i>		
Nominal rate	R^n	4.08%
Real rate	R	2.03%
Equity return	R^{eq}	3.26%
Long bond return	R^L	2.96%
Equity spread	$R^{eq} - R$	121 bp
Bond spread	$R^L - R$	92 bp
<i>Financial Intermediaries</i>		
Leverage ratio	ϕ_{fi}	3.55
Net worth	N_{fi}	29.66
Long bond price	Q^L	16.04

3.3 Key Parameters for Results

Three sets of parameters are particularly important for the profitability and welfare results.

First, the *market participation share* ψ_u controls the degree of Wallace neutrality. As $\psi_u \rightarrow 1$, more households can rebalance their portfolios in response to central bank purchases, dampening the real effects of QE. Conversely, lower ψ_u strengthens QE transmission

by increasing the fraction of households who cannot offset the central bank’s market intervention. Section 5 examines $\psi_u \in \{0.3, 0.5, 0.7\}$.

Second, the *relative divertability* λ_{eq}/λ_L determines the ratio of equity to bond spreads and the relative effectiveness of equity versus bond purchases in relaxing the intermediary leverage constraint. Higher λ_{eq}/λ_L widens the equity spread relative to bonds and strengthens the case for equity QE as a leverage-relaxing tool. It also affects profitability through the spread channel: higher equity spreads mean the central bank earns a higher return on equity holdings.

Third, the *bond duration* χ controls the interest rate sensitivity of bond prices and therefore the magnitude of capital losses when rates normalise. Higher duration unambiguously increases the buy-high-sell-low problem for bond QE, strengthening the relative profitability advantage of equity QE.

3.4 Bayesian Estimation

To discipline the model’s dynamics, we estimate 16 parameters using Bayesian methods. The estimation is conducted on a version of the model without QE or the ZLB constraint (a standard linear New Keynesian model with financial intermediaries), which shares the same steady state and linearised dynamics as the full model outside of ZLB episodes.

3.4.1 Data

We use seven US macroeconomic time series at quarterly frequency from 1984Q1 to 2025Q3 ($T = 167$ observations). The observables are: real GDP, real personal consumption expenditure, real gross private domestic investment, the GDP deflator inflation rate, the effective federal funds rate, real compensation per hour, and total hours worked. All series are obtained from FRED (Federal Reserve Bank of St. Louis).

Each series is expressed in logs (except the interest rate, which is expressed in levels) and detrended using the Hodrick-Prescott filter with smoothing parameter $\lambda = 1600$. The observables enter the likelihood as log deviations from their HP-filtered trends. To avoid stochastic singularity (seven observables but only seven structural shocks), we include small i.i.d. measurement errors on each observable with standard deviation fixed at 0.001.

3.4.2 Prior Specification

Table 3 reports the prior and posterior distributions for all estimated parameters. We estimate three structural parameters ($\phi_\pi, \rho_r, \sigma_c$), six shock persistence parameters ($\rho_Z, \rho_A, \rho_{BL}, \rho_G, \rho_{rp}, \rho_{inv}$), and seven shock standard deviations ($\sigma_{\varepsilon Z}, \sigma_{\varepsilon A}, \sigma_{\varepsilon M}, \sigma_{\varepsilon BL}, \sigma_{\varepsilon G}, \sigma_{\varepsilon RP}, \sigma_{\varepsilon Inv}$).

Prior distributions are standard in the DSGE literature. The inflation response ϕ_π has a Gamma prior centred at 1.5, ensuring the Taylor principle is satisfied. Persistence parameters have Beta priors on $(0, 1)$. Shock standard deviations have Inverse Gamma priors, which are positive-valued and have fat tails to accommodate the wide range of plausible shock volatilities. All remaining parameters—including the financial intermediary parameters, holding costs, and fiscal rule coefficients—are calibrated as described above.

3.4.3 Posterior Results

Estimation uses the Random Walk Metropolis-Hastings algorithm with two parallel chains of 100,000 draws each, discarding the first 50 percent as burn-in. The posterior mode is found using the `csminwel` optimiser, and the proposal covariance is set to the inverse of the posterior Hessian evaluated at the mode.

Several features of the posterior are noteworthy. First, the estimated inflation response $\hat{\phi}_\pi = 2.62$ is substantially above its prior mean of 1.5, consistent with an aggressive anti-inflation stance in the post-Volcker sample. Second, interest rate smoothing is high ($\hat{\rho}_r = 0.91$), reflecting the Federal Reserve’s well-documented gradualism in adjusting the policy rate. Third, risk aversion $\hat{\sigma}_c = 2.71$ is higher than the standard calibration of 1.5 but well within the range of estimates in the literature.

Among the shock processes, two results stand out. The preference shock is near-unit-root ($\hat{\rho}_A = 0.997$), indicating that demand disturbances are highly persistent—consistent with the slow recovery from the Great Recession and the protracted period of low interest rates that followed. The investment efficiency shock has low persistence ($\hat{\rho}_{inv} = 0.24$) but large volatility ($\hat{\sigma}_{inv} = 0.136$), suggesting that investment disruptions are sharp but transitory.

The log marginal data density (Laplace approximation) is 4090.7, providing a benchmark for model comparison. We use the estimated parameter values (posterior modes) in all subsequent experiments, while maintaining the calibrated values for parameters not included in the estimation.

4 Results

We solve the model using Dynare’s OccBin toolkit ([Guerrieri and Iacoviello, 2015](#)), which handles the ZLB as an occasionally binding constraint via piecewise-linear methods. For each experiment, we compute a 200-period simulation path under a specific shock realisation, allowing the solver to endogenously determine when and for how long the ZLB binds. We consider four QE regimes (no QE, bond-only, equity-only, mixed) under seven shocks (risk premium, preference, financial, TFP, monetary policy, government spending, investment

Table 3: Bayesian Estimation Results

Parameter	Description	Prior	Prior Mean	Post. Mean	90% HPD
<i>Structural Parameters</i>					
ϕ_π	Inflation response	$\Gamma(1.5, 0.25)$	1.500	2.580	[2.182, 2.953]
ρ_r	Interest rate smoothing	$B(0.75, 0.10)$	0.750	0.909	[0.893, 0.925]
σ_c	Risk aversion	$\Gamma(1.5, 0.25)$	1.500	2.689	[2.068, 3.338]
<i>Shock Persistence</i>					
ρ_Z	TFP	$B(0.90, 0.05)$	0.900	0.904	[0.861, 0.949]
ρ_A	Preference	$B(0.80, 0.10)$	0.800	0.997	[0.996, 0.999]
ρ_{BL}	Bond supply	$B(0.90, 0.05)$	0.900	0.967	[0.948, 0.987]
ρ_G	Gov. spending	$B(0.90, 0.05)$	0.900	0.860	[0.814, 0.906]
ρ_{rp}	Risk premium	$B(0.80, 0.10)$	0.800	0.847	[0.816, 0.876]
ρ_{inv}	Inv. efficiency	$B(0.80, 0.10)$	0.800	0.245	[0.184, 0.311]
<i>Shock Standard Deviations</i>					
σ_Z	TFP	$IG(0.01, 0.05)$	0.010	0.005	[0.004, 0.005]
σ_A	Preference	$IG(0.01, 0.05)$	0.010	0.066	[0.057, 0.075]
σ_M	Monetary policy	$IG(0.005, 0.05)$	0.005	0.001	[0.001, 0.002]
σ_{BL}	Bond supply	$IG(0.01, 0.05)$	0.010	0.046	[0.039, 0.054]
σ_G	Gov. spending	$IG(0.01, 0.05)$	0.010	0.023	[0.021, 0.025]
σ_{RP}	Risk premium	$IG(0.01, 0.05)$	0.010	0.011	[0.008, 0.014]
σ_{Inv}	Inv. efficiency	$IG(0.01, 0.05)$	0.010	0.138	[0.124, 0.151]
Log marginal data density (Laplace): 4090.7; Modified Harmonic Mean: 4091.8					

Notes: Γ , B , and IG denote Gamma, Beta, and Inverse Gamma prior distributions. Prior columns report the prior mean and standard deviation. Posterior mean and 90% highest posterior density (HPD) intervals from Metropolis–Hastings with $2 \times 100,000$ draws (acceptance rate: 17%). Sample: US data, 1984Q1–2025Q3 ($T = 167$).

efficiency) and two unwinding strategies (passive and active), yielding 56 experiments in total. Of the seven shocks, only the risk premium shock generates a ZLB episode in the piecewise-linear approximation. We focus the analysis on this shock, which provides the most natural financial-crisis scenario motivating QE. All model parameters are set to their posterior mean estimates from the Bayesian estimation (Table 3).

4.1 Baseline: Risk Premium Shock

The risk premium shock $\varepsilon^{RP} = 0.042$ generates a financial-crisis-type disturbance: the risk premium shifter $v_{rp,t}$ rises, increasing the required return on equity and widening credit spreads. Output falls by 3.4 percent at the trough under no QE, and the nominal rate is constrained at the ZLB for 6 quarters. With QE active, the ZLB duration extends to 8–9 quarters and the output trough deepens to 4.2–5.0 percent, reflecting the fiscal cost of QE losses feeding back through higher labor taxes.

4.1.1 Impulse Response Functions

Figure 1 displays impulse responses under passive unwinding for the four QE regimes. Several features are noteworthy. Output and consumption decline sharply on impact, with the fall in investment even more pronounced as the elevated risk premium raises the cost of capital. The nominal rate hits the ZLB within two quarters and remains there for 6 quarters (without QE) or 8–9 quarters (with QE). Inflation falls modestly, reflecting the demand-side nature of the shock.

Crucially, asset prices move in opposite directions. Bond prices Q^L rise during the ZLB episode as yields are compressed to near-zero levels. Equity prices Q^K fall below steady state as the elevated risk premium depresses equity valuations. This price asymmetry is the fundamental driver of the profitability results. The last panel shows the labor tax rate τ_t^L , which rises as the fiscal rule responds to reduced central bank remittances and debt deviations. The tax path differs across QE regimes: bond QE, which generates the largest losses, induces the highest tax rate, while equity QE—particularly with active unwinding—produces the smallest fiscal burden.

The consolidated government debt panel shows that $Q_t^L(B_t^L - B_t^{CB})$ —the market value of bonds held by the private sector—rises during the recession. Bond-heavy programs produce more pronounced reductions in privately held debt, while equity QE leaves consolidated debt largely unchanged.

Figure 1: IRFs to Risk Premium Shock ($\varepsilon_{RP} = 0.042$), Passive Unwind

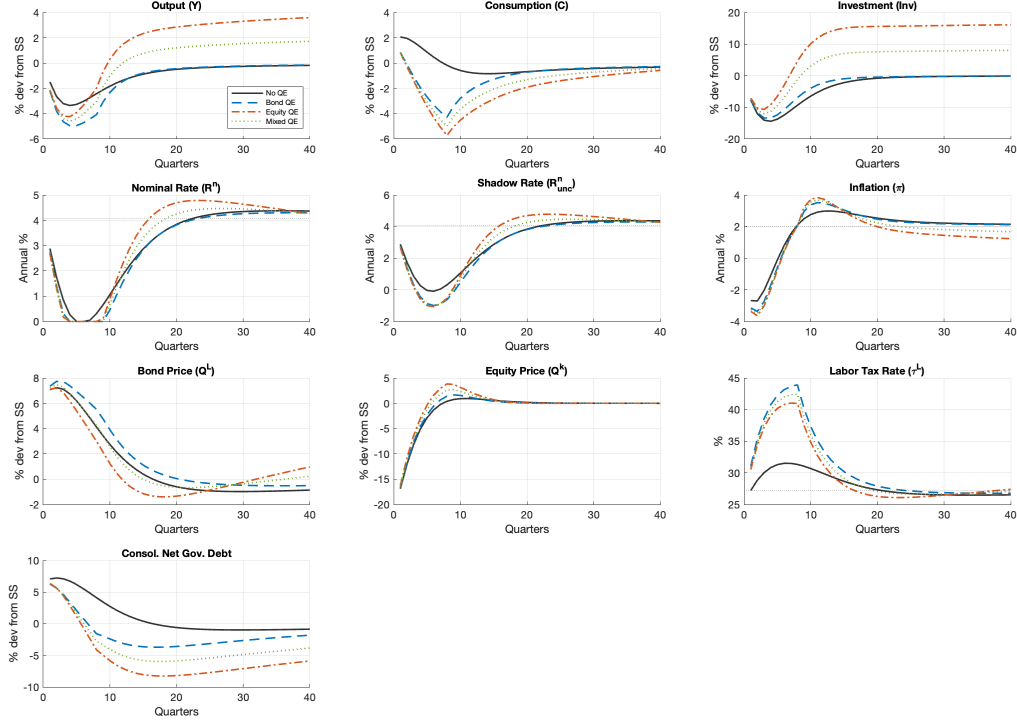


Figure 1: Impulse responses to a risk premium shock ($\varepsilon^{RP} = 0.042$) under passive unwinding. Panels show output, consumption, investment, nominal rate, shadow rate, inflation, bond price (Q^L), equity price (Q^K), labor tax rate (τ^L), and consolidated government debt ($Q_t^L(B_t^L - B_t^{CB})$). Rate variables are shown as annualized levels; all other variables are in percent deviation from steady state.

4.1.2 Central Bank Balance Sheet and Profits

Figure 2 shows the central bank balance sheet dynamics in detail. Under bond-only QE, the central bank accumulates bonds during the ZLB period and holds them as they slowly decay (passive unwind) or actively sells after liftoff (active unwind). Under equity-only QE, the central bank buys equities at depressed prices and either retains them indefinitely or sells gradually during the recovery.

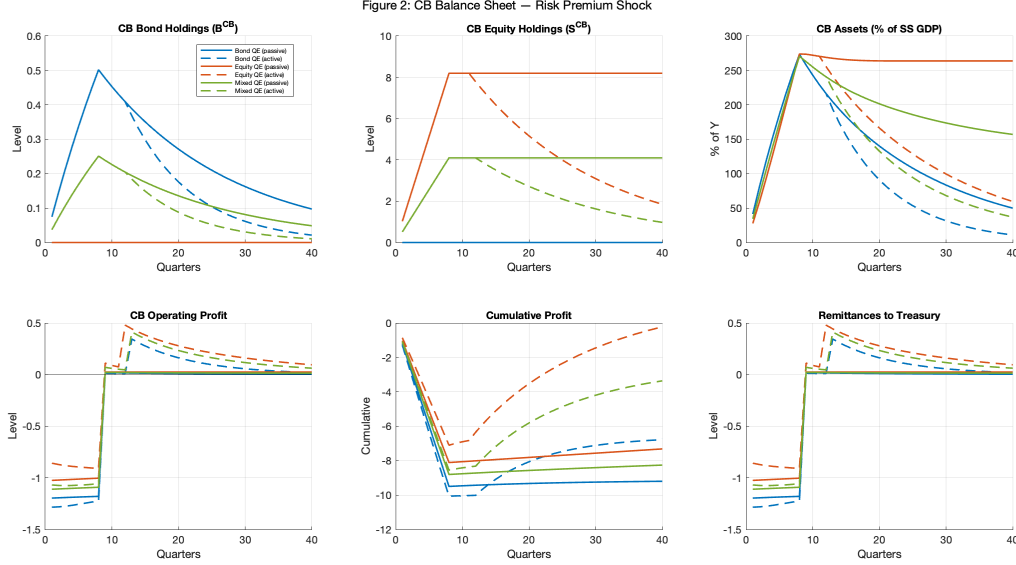


Figure 2: Central bank balance sheet dynamics under the risk premium shock. Top panels: bond and equity holdings by regime. Bottom panels: total balance sheet as a fraction of GDP and cumulative profits.

Figure 3 decomposes central bank profits into their components. Bond QE generates losses through two channels: the capital loss channel (bond prices fall as rates normalise) and the carry channel (the central bank pays rising reserve interest while earning fixed nominal coupons on its bond portfolio). Equity QE generates partially offsetting dynamics: while funding costs also rise, the capital gains on equities (which appreciate during the recovery) partially offset the carry cost, and dividend income (Z_k) is procyclical.

Table 5 presents the present value of central bank profits across all experiments. Under the risk premium shock with passive unwinding, bond QE generates a PV loss of -9.008 , while equity QE generates a substantially smaller loss of -5.034 —a 44 percent improvement. With active unwinding, the results are even more striking: *equity QE generates a net profit* of $+0.584$, while bond QE still loses -6.770 . This is the paper's central quantitative result: with active unwinding, equity QE turns the central bank balance sheet from a fiscal liability

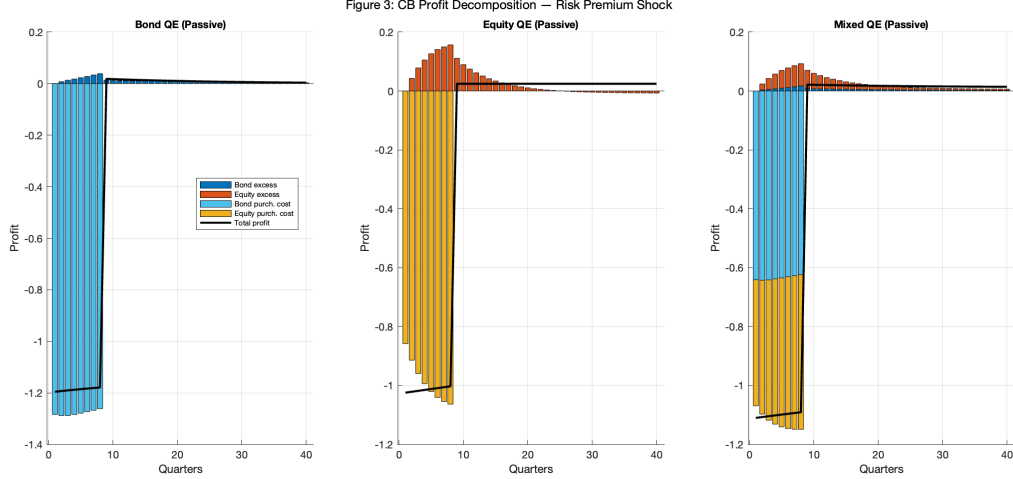


Figure 3: Decomposition of central bank profits under the risk premium shock with passive unwinding. Panels show the income component (coupons/dividends minus reserve interest) and the capital gain/loss component for each QE regime.

into a fiscal asset.

The intuition for why active unwinding favours equity QE is straightforward. Active unwinding crystallises capital gains on equities (which have recovered in price) while crystallising capital losses on bonds (whose prices have fallen as rates normalise). Under passive unwinding, these valuation effects remain unrealised, and the profit difference is driven primarily by the flow income channel (dividends versus coupons relative to reserve costs).

4.2 Other Shocks

Table 4 reports the ZLB duration and output trough for each shock type. Six of the seven shocks do not generate ZLB episodes in the piecewise-linear approximation. The preference shock ($\varepsilon^A = 0.50$), despite generating a substantial output decline of 6.0 percent, does not push the nominal rate to the ZLB because the estimated near-unit-root persistence ($\rho_A = 0.997$) produces a slow-moving demand shift rather than an acute recession. The financial shock has near-zero real effects due to the linear holding cost structure. The TFP shock reduces output but generates inflationary pressure that keeps the nominal rate above zero. The monetary policy, government spending, and investment efficiency shocks all produce modest output declines at the calibrated sizes without ZLB binding.

Since the QE rules are only activated when the ZLB binds, all QE regimes produce identical zero outcomes for these shocks: $PV^{CB} = 0$, no balance sheet accumulation, no profit or loss. This result highlights that the profitability comparison is relevant specifically

for financial-crisis-type episodes that drive the economy to the ZLB.

Figure 4 shows the impulse responses under the preference shock for comparison. Despite the deep recession, the shadow rate remains above the ZLB throughout, and QE is never activated. This illustrates how the nature of the shock—not just its severity—determines whether QE is deployed.

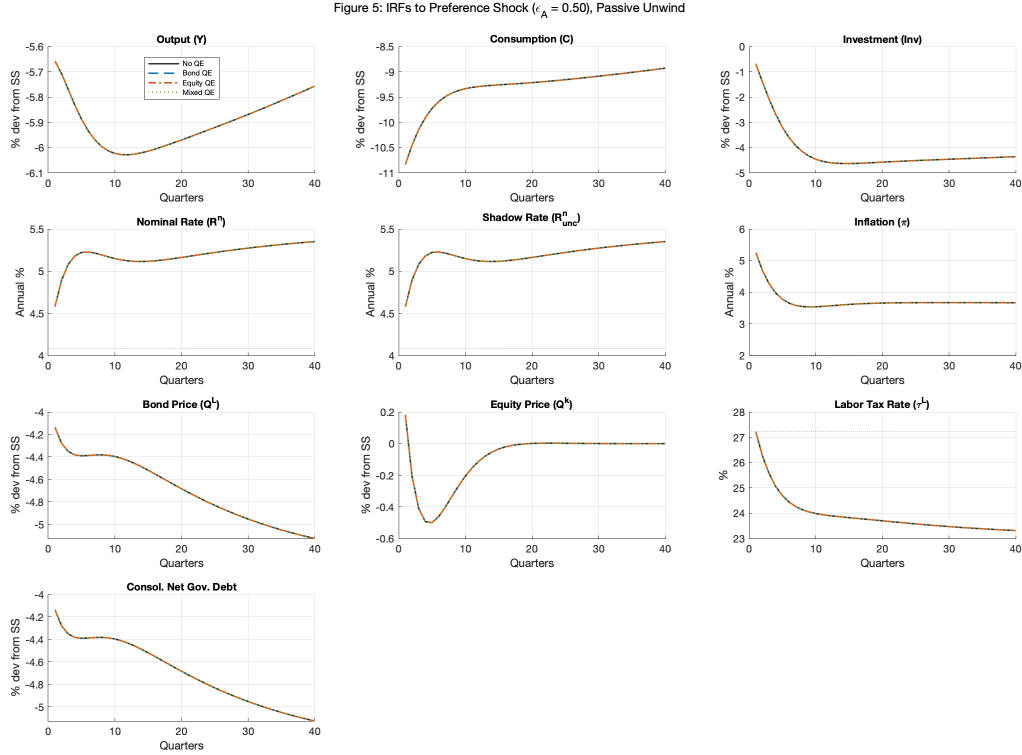


Figure 4: Impulse responses to a preference shock ($\epsilon^A = 0.50$) under passive unwinding. Despite a 6% output decline, the shadow rate remains above the ZLB and QE is never activated. All QE regimes produce identical paths.

4.3 Profitability Summary

Figure 5 and Table 5 present the complete profitability results. The main findings can be summarised in four points.

First, *equity QE dominates bond QE in profitability*. The PV loss under equity QE is 44 percent smaller than bond QE with passive unwinding. With active unwinding, equity QE generates a *profit* of +0.584 while bond QE loses -6.770 .

Second, *active unwinding transforms equity QE from a loss-making to a profit-making operation*. Under passive unwinding, equity QE still loses -5.034 , but switching to active

Table 4: ZLB Duration and Output Trough by Shock Type

Shock	Shock Size	ZLB Duration	Output Trough	Shadow Rate Min
Risk Premium	$\varepsilon^{RP} = 0.04$	6 quarters	-3.35%	0.9998
Preference	$\varepsilon^A = 0.50$	0 quarters	-6.03%	1.0113
Financial	$\varepsilon^{Fin} = 0.10$	0 quarters	-0.00%	1.0101
TFP	$\varepsilon^Z = -0.03$	0 quarters	-1.45%	1.0101
Monetary	$\varepsilon^M = 0.01$	0 quarters	-1.09%	1.0099
Gov. Spending	$\varepsilon^G = 0.05$	0 quarters	-0.75%	1.0092
Inv. Efficiency	$\varepsilon^{Inv} = 0.10$	0 quarters	-0.45%	1.0093

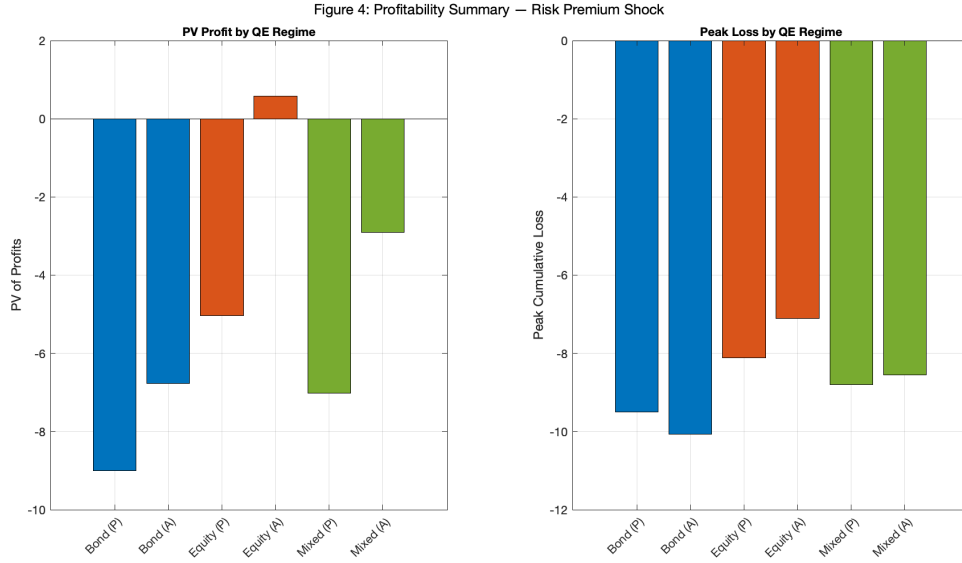


Figure 5: Present value of central bank profits across QE regimes and unwinding strategies under the risk premium shock. (P) = passive, (A) = active unwinding.

Table 5: Central Bank Profitability Across QE Regimes

Shock	QE Regime	Passive Unwind			Active Unwind		
		PV Prof	Peak Loss	Peak BS	PV Prof	Peak Loss	Peak BS
Risk Premium	No QE	-0.000	-0.000	0.0%	-0.000	-0.000	0.0%
	Bond	-9.008	-9.496	273.4%	-6.770	-10.064	273.4%
	Equity	-5.034	-8.108	273.9%	0.584	-7.104	273.9%
	Mixed	-7.021	-8.802	270.4%	-2.905	-8.548	270.4%
Preference	No QE	0.000	-0.000	0.0%	0.000	-0.000	0.0%
	Bond	0.000	-0.000	0.0%	0.000	-0.000	0.0%
	Equity	0.000	-0.000	0.0%	0.000	-0.000	0.0%
	Mixed	0.000	-0.000	0.0%	0.000	-0.000	0.0%
Financial	No QE	0.000	0.000	0.0%	0.000	0.000	0.0%
	Bond	0.000	0.000	0.0%	0.000	0.000	0.0%
	Equity	0.000	0.000	0.0%	0.000	0.000	0.0%
	Mixed	0.000	0.000	0.0%	0.000	0.000	0.0%
TFP	No QE	0.000	-0.000	0.0%	0.000	-0.000	0.0%
	Bond	0.000	-0.000	0.0%	0.000	-0.000	0.0%
	Equity	0.000	-0.000	0.0%	0.000	-0.000	0.0%
	Mixed	0.000	-0.000	0.0%	0.000	-0.000	0.0%
Monetary	No QE	-0.000	-0.000	0.0%	-0.000	-0.000	0.0%
	Bond	-0.000	-0.000	0.0%	-0.000	-0.000	0.0%
	Equity	-0.000	-0.000	0.0%	-0.000	-0.000	0.0%
	Mixed	-0.000	-0.000	0.0%	-0.000	-0.000	0.0%
Gov. Spending	No QE	-0.000	-0.000	0.0%	-0.000	-0.000	0.0%
	Bond	-0.000	-0.000	0.0%	-0.000	-0.000	0.0%
	Equity	-0.000	-0.000	0.0%	-0.000	-0.000	0.0%
	Mixed	-0.000	-0.000	0.0%	-0.000	-0.000	0.0%
Inv. Efficiency	No QE	-0.000	-0.000	0.0%	-0.000	-0.000	0.0%
	Bond	-0.000	-0.000	0.0%	-0.000	-0.000	0.0%
	Equity	-0.000	-0.000	0.0%	-0.000	-0.000	0.0%
	Mixed	-0.000	-0.000	0.0%	-0.000	-0.000	0.0%

Notes: PV Prof = present value of CB profits discounted at SS real rate. Peak Loss = minimum cumulative profit. Peak BS = peak CB balance sheet as % of quarterly GDP.

unwinding produces a net gain of +0.584—a swing of 5.618. Bond QE improves by a smaller amount (2.238, or 25 percent). The asymmetry arises because active unwinding allows the central bank to crystallise equity capital gains, whereas bond capital losses are crystallised regardless of the unwinding strategy.

Third, *QE itself deepens the recession through the fiscal channel*. The output trough under no QE is −3.35 percent, but worsens to −4.95 percent with bond QE, −4.23 percent with equity QE, and −4.59 percent with mixed QE. This occurs because QE losses feed back through higher labor taxes, which distort labor supply and reduce output. Equity QE, which generates smaller losses, produces a smaller contractionary fiscal feedback than bond QE.

Fourth, the mixed QE regime performs intermediate between bond and equity QE. Under passive unwinding, mixed QE loses −7.021; under active unwinding, it loses −2.905. Adding bonds to an equity portfolio worsens the profitability outcome, while adding equities to a bond portfolio improves it—but not enough to offset the bond drag.

4.3.1 Sensitivity to ZLB Duration

Figure 6 examines how the profitability ranking varies with the severity of the risk premium shock. We consider three shock sizes— $\varepsilon^{RP} \in \{0.042, 0.050, 0.060\}$ —producing ZLB episodes of increasing duration. Each row of the figure shows the output path, nominal rate, cumulative central bank profit, and consolidated government debt for the four QE regimes under passive unwinding.

The qualitative ranking is stable across shock sizes: equity QE consistently produces smaller cumulative losses than bond QE, while mixed QE performs intermediate. Larger shocks amplify the absolute magnitude of all losses but preserve the equity advantage. The consolidated debt panels show that larger shocks produce greater reductions in privately held debt under bond-heavy programs, as the central bank absorbs a larger share of outstanding bonds.

4.4 Unwinding Strategies

Figure 7 compares passive and active unwinding for each QE regime under the risk premium shock. Two findings stand out.

First, the unwinding strategy matters far more for equity QE than for bond QE. The PV improvement from switching to active unwinding is 5.618 for equity QE (turning a loss into a profit) but only 2.238 for bond QE. This is because active equity unwinding crystallises the large capital gains that accumulate during the recovery, whereas passive equity unwinding leaves these gains unrealised indefinitely.

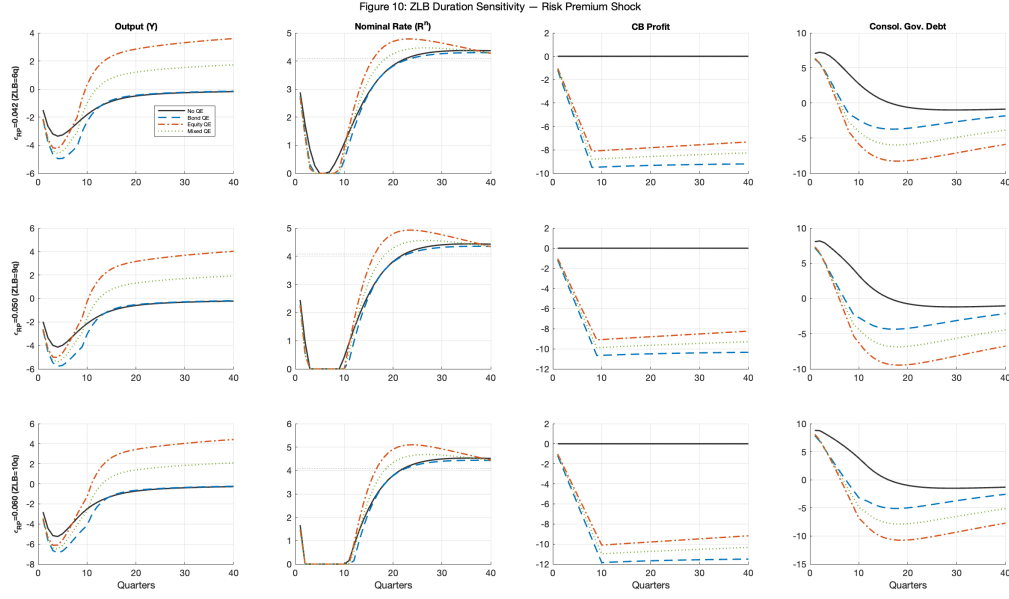


Figure 6: ZLB duration sensitivity. Each row corresponds to a different risk premium shock size. Columns show output, nominal rate, cumulative CB profit, and consolidated government debt. Passive unwinding in all cases.

Second, passive unwinding is more costly for bonds than for equities, even in relative terms. Under bond QE, the central bank earns below-market coupons for an extended period while paying rising interest on reserves. This carry cost accumulates slowly but persistently. Under equity QE, dividend income (Z_k) recovers alongside the economy, partially offsetting the rising funding cost even without active sales.

4.5 Welfare

Table 6 reports welfare effects of QE under the risk premium shock, expressed in consumption-equivalent variation (CEV). We compute CEV separately for unrestricted and restricted households, as well as the population-weighted aggregate. Since only the risk premium shock activates QE, all other shocks produce zero welfare differences across QE regimes. Passive and active unwinding produce identical welfare because the active unwind is implemented as a post-processing adjustment to the central bank balance sheet, leaving equilibrium consumption paths unchanged.

The distributional pattern is systematic and economically interpretable. Two opposing channels drive welfare across household types. First, the *portfolio channel*: central bank purchases reduce the supply of assets available to unrestricted households, raising equilib-

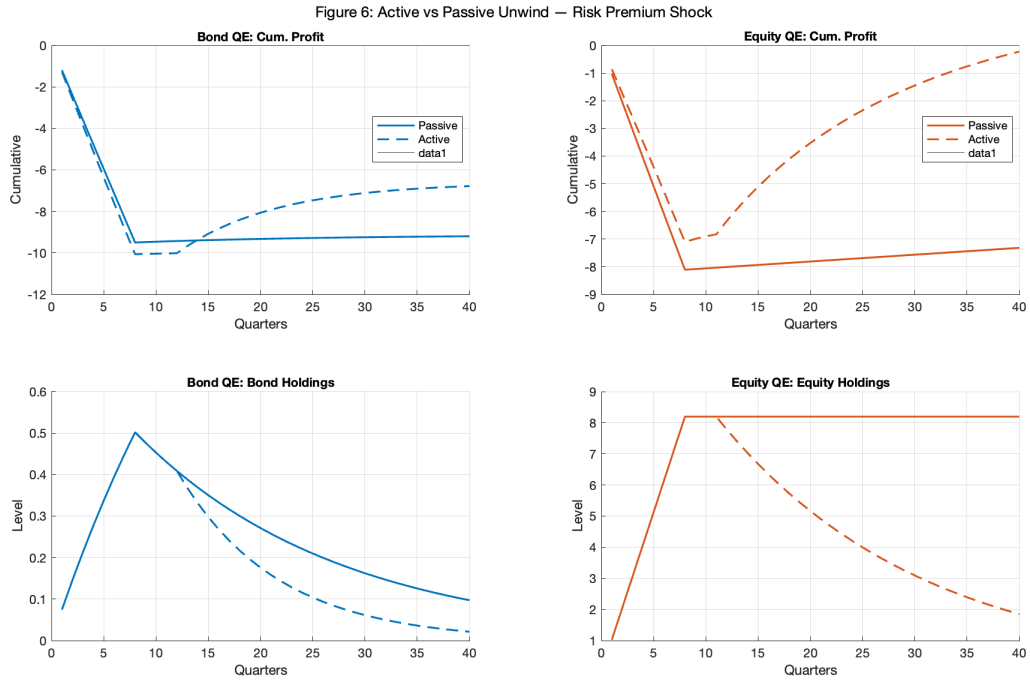


Figure 7: Comparison of unwinding strategies under the risk premium shock. Left panels: cumulative profits for bond and equity QE. Right panels: balance sheet holdings. Solid lines: passive unwinding; dashed lines: active unwinding.

Table 6: Welfare Gains from QE (CEV, %)

Shock	QE Regime	Passive Unwind			Active Unwind		
		UHH	RHH	Aggregate	UHH	RHH	Aggregate
Risk Premium	Bond	0.1182	-0.8295	-0.3556	0.1182	-0.8295	-0.3556
Risk Premium	Equity	-0.2008	0.0486	-0.0761	-0.2008	0.0486	-0.0761
Risk Premium	Mixed	-0.0279	-0.3896	-0.2087	-0.0279	-0.3896	-0.2087
Preference	Bond	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Equity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Mixed	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Financial	Bond	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Equity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Mixed	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
TFP	Bond	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Equity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Mixed	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Monetary	Bond	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Equity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Mixed	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Gov. Spending	Bond	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Equity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Mixed	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Inv. Efficiency	Bond	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Equity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	Mixed	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Notes: Welfare gains expressed as consumption-equivalent variation relative to the no-QE baseline for each shock. Positive values indicate welfare improvement from QE.

rium asset prices and benefiting existing holders. This explains why bond QE produces a welfare *gain* for unrestricted households (+0.12% CEV) despite generating aggregate losses. Second, the *fiscal channel*: QE losses feed through to higher labor taxes τ_t^L , which fall disproportionately on hand-to-mouth consumers who cannot smooth the burden through financial markets. Bond QE, which generates the largest losses, imposes a -0.83% CEV loss on restricted households. Equity QE, by generating smaller losses, produces a slight welfare *gain* for restricted households (+0.05% CEV) relative to no QE—the smaller fiscal burden more than offsets the absence of portfolio channel benefits.

In aggregate (population-weighted), equity QE produces the smallest welfare loss (-0.08% CEV), followed by mixed (-0.21%) and bond QE (-0.36%). The welfare ranking mirrors the profitability ranking, confirming that the fiscal channel is the primary mechanism through which QE composition affects household welfare.

4.6 The Role of Household Heterogeneity

The two-household structure is essential for three aspects of the results. First, restricted households cannot rebalance their portfolios in response to central bank purchases, ensuring that QE creates genuine asset scarcity and affects equilibrium prices. In a representative-agent version ($\psi_u = 1$), this channel would be attenuated (though not eliminated, since the intermediary leverage constraint still binds).

Second, the fiscal channel operates asymmetrically across household types. Changes in the labor tax rate τ_t^L affect restricted households directly through the after-tax wage $W_t(1 - \tau_t^L)$ in their budget constraint, while unrestricted households can partially smooth the impact through portfolio rebalancing. This makes the welfare implications of central bank profitability sharper: the distortionary cost of higher taxes falls disproportionately on those least able to smooth them.

Third, the heterogeneity creates a portfolio composition effect. When the central bank purchases equities, it reduces the supply available to unrestricted households and intermediaries. Unrestricted households bear part of this adjustment through their holding cost, while intermediaries bear it through the leverage constraint. The relative burden depends on ψ_u and the holding cost parameters—a relationship explored in the sensitivity analysis.

5 Sensitivity Analysis

We examine the robustness of the main results to variation in four key parameters: the household market participation share, the relative asset divertability, bond duration, and

the QE scale and unwinding speed. Table 7 summarises the PV of central bank profits across parameter values, and Figure 8 visualises the sensitivity patterns.

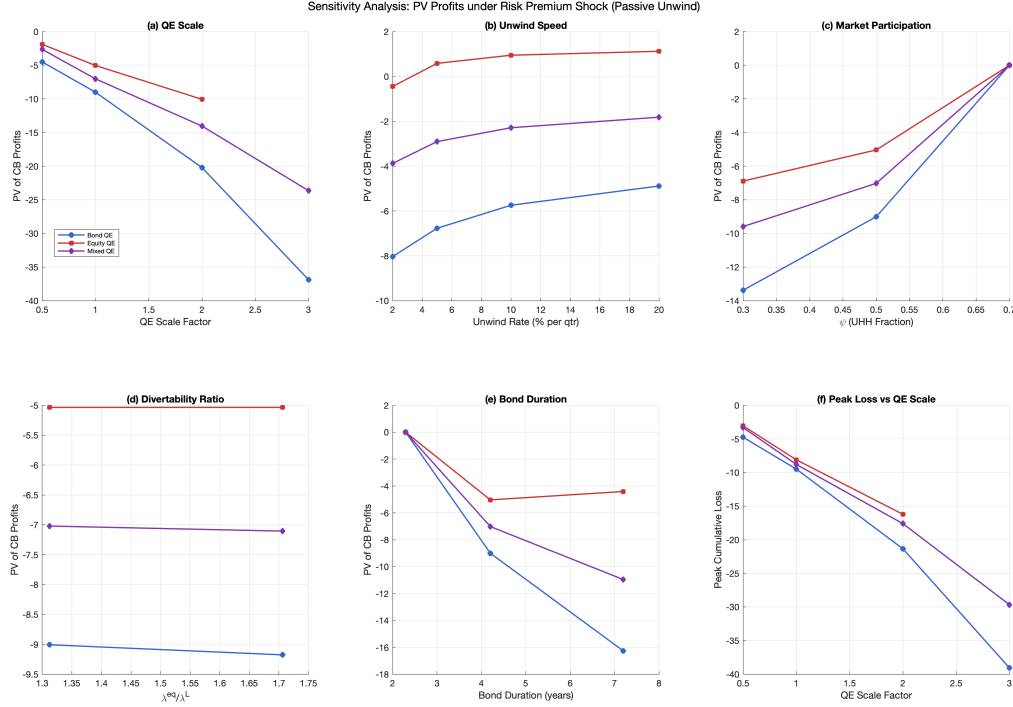


Figure 8: Sensitivity of PV profits to key parameters under the risk premium shock with passive unwinding (except panel (b) which uses active unwinding). Each line represents a QE regime: bond-only (blue circles), equity-only (red squares), and mixed (purple diamonds). Panel (f) shows peak cumulative losses versus QE scale.

5.1 Household Market Participation (ψ_u)

The fraction of unrestricted households ψ_u controls the degree to which private portfolio rebalancing offsets central bank purchases. We vary $\psi_u \in \{0.3, 0.5, 0.7\}$, re-solving the steady state for each value.

As ψ_u increases toward unity, QE becomes less effective: more households can rebalance their portfolios in response to central bank purchases, dampening the asset-price and spread effects. The profitability of all QE regimes declines in absolute terms because the central bank accumulates a smaller effective position (more of its purchases are offset by private rebalancing).

The central finding is that the *relative* advantage of equity QE over bond QE is robust across values of ψ_u . Equity QE generates smaller losses than bond QE at all participation

Table 7: Sensitivity of PV Profits to Key Parameters

Parameter	Value	Bond QE	Equity QE	Mixed QE
<i>Panel A: QE Scale Factor</i>				
	0.5×	-4.504	-1.894	-2.644
	1.0×	-9.008	-5.034	-7.021
	2.0×	-20.217	-10.069	-14.042
	3.0×	-36.881	—	-23.645
<i>Panel B: Unwind Speed (ρ^{unwind}, active)</i>				
	2% / qtr	-8.031	-0.447	-3.877
	5% / qtr	-6.770	0.584	-2.905
	10% / qtr	-5.744	0.948	-2.286
	20% / qtr	-4.885	1.122	-1.817
<i>Panel C: Market Participation (ψ)</i>				
	0.30	-13.378	-6.889	-9.592
	0.50	-9.008	-5.034	-7.021
	0.70	0.000	0.000	0.000
<i>Panel D: Divertability Ratio (λ^{eq}/λ^L)</i>				
	1.05	—	—	—
	1.31	-9.008	-5.034	-7.021
	1.71	-9.177	-5.034	-7.106
<i>Panel E: Bond Duration (χ)</i>				
	$\chi=0.900$ (2.3 yr)	0.000	0.000	0.000
	$\chi=\mathbf{0.950}$ (4.2 yr)	-9.008	-5.034	-7.021
	$\chi=0.975$ (7.2 yr)	-16.265	-4.412	-10.960

Notes: PV of CB profits under risk premium shock ($\varepsilon^{RP} = 0.042$) with passive unwind (except Panel B which uses active unwind). Baseline values in bold. — indicates non-convergence.

rates. The intuition is that the buy-high-sell-low mechanism for bonds and the buy-low-sell-high mechanism for equities both depend on asset-price dynamics at the ZLB—dynamics that are driven by the shock itself, not by the degree of market participation. Higher ψ_u attenuates the *level* of profits and losses (by making QE less potent overall) but does not reverse the *ranking*.

The welfare implications are more sensitive to ψ_u . As ψ_u rises, the fiscal channel weakens because a smaller fraction of the population is hand-to-mouth. The distributional pattern—equity QE benefiting restricted households—persists but becomes less pronounced.

5.2 Asset-Specific Divertability (λ_{eq}/λ_L)

The ratio λ_{eq}/λ_L determines the relative credit spreads on equities and bonds in steady state (the spread ratio equals $\gamma_S/\gamma_B = \lambda_{eq}/\lambda_L$ by construction). We vary this ratio by adjusting λ_{eq} while holding λ_L fixed, and recalibrating γ_B to maintain consistency.

Higher λ_{eq}/λ_L (greater relative equity divertability) widens equity spreads, which has two effects on profitability. First, it increases the central bank’s carry income on equity holdings (since dividends and equity returns are higher relative to the deposit rate). Second, it makes equity QE more powerful per dollar in relaxing the intermediary leverage constraint, potentially generating larger macroeconomic effects and larger asset-price movements.

When λ_{eq}/λ_L is reduced toward unity (equities and bonds equally divertable), the profitability advantage of equity QE narrows but does not disappear entirely. Even with identical credit spreads, the fundamental price-dynamics asymmetry persists: bond prices are high at the ZLB while equity prices are low. The spread difference amplifies this asymmetry but is not its sole source.

5.3 Bond Duration (χ)

Bond duration, controlled by the coupon decay rate χ , affects the interest rate sensitivity of bond prices and therefore the magnitude of the buy-high-sell-low problem for bond QE. We consider $\chi \in \{0.90, 0.95, 0.975\}$, corresponding to durations of approximately 2.5, 3.8, and 7.5 years.

Higher duration unambiguously increases bond QE losses. With $\chi = 0.975$ (7.5-year duration), the PV loss under bond QE with passive unwinding rises substantially, as the larger price swings in long bonds amplify both the mark-to-market losses during the normalisation phase and the opportunity cost of holding below-market coupons.

Equity QE profitability is largely unaffected by bond duration, since equity price dynamics are driven by investment fundamentals (Tobin’s Q), not by the term structure. The

profitability advantage of equity QE therefore *increases* with bond duration. This finding is relevant for central banks whose QE portfolios are concentrated in very long-dated bonds (e.g., the Bank of England’s gilt portfolio, which had a weighted average maturity exceeding 15 years during QE).

5.4 QE Scale and Unwinding Speed

We examine whether the profitability ranking is robust to the scale of QE purchases and the speed of unwinding.

QE scale. Doubling the purchase intensities (Φ_B and Φ_E) roughly doubles the peak balance sheet and approximately doubles the PV losses for each regime. The profitability *ratio* (equity QE loss / bond QE loss) remains approximately constant, confirming that the ranking is not an artefact of the particular calibration of purchase intensities. There are no strong nonlinearities within the range considered, reflecting the (piecewise) linear nature of the model solution.

Unwinding speed. Faster active unwinding (higher ρ_B^{unwind} and ρ_E^{unwind}) sharpens the profitability advantage of equity QE. With very rapid unwinding (e.g., selling 20 percent of the portfolio per quarter), the central bank crystallises equity gains quickly after the price recovery, while crystallising bond losses equally quickly. The optimal unwinding speed from a pure profitability standpoint is an increasing function of how quickly asset prices normalise relative to the unwinding trajectory.

Slower unwinding ($\rho^{unwind} = 0.02$ per quarter) converges toward the passive unwinding result, as expected. The active-passive gap narrows but remains positive for equity QE, confirming that even modest active unwinding substantially improves the equity QE profit outcome.

5.5 Fiscal Response

The distortionary labor tax creates a direct channel from central bank profitability to real outcomes. The welfare results are sensitive to the fiscal rule specification, particularly the tax smoothing parameter ρ_τ and the remittance passthrough ϕ_{Rem} .

With lower tax smoothing ($\rho_\tau = 0.3$), the labor tax rate τ_t^L adjusts more rapidly to central bank losses, creating sharper distortions to the after-tax wage $W_t(1 - \tau_t^L)$ and larger consumption fluctuations for restricted households. This amplifies the welfare advantage of equity QE (which generates smaller losses and hence smaller tax distortions) while increasing the welfare cost of bond QE.

With higher tax smoothing ($\rho_\tau = 0.9$), the fiscal burden is spread over a longer horizon, dampening the immediate labor supply distortion. The distributional pattern persists but is attenuated: the tax wedge adjusts slowly, partially insulating restricted households from the short-run consequences, and the welfare difference between equity and bond QE shrinks.

Higher remittance passthrough ($\phi_{Rem} = 0.20$) strengthens the direct channel from central bank profits to the tax rate, amplifying the differences across QE regimes. The labor supply distortion—and hence the output trough differential—widens with ϕ_{Rem} , as central bank losses translate more aggressively into higher tax rates.

These results underscore that the welfare channel depends critically on institutional features of the fiscal-monetary interface—how quickly and transparently central bank losses are passed through to the fiscal authority, and how the fiscal authority distributes the burden through distortionary taxation.

5.6 Role of the Zero Lower Bound

A natural question is whether the profitability advantage of equity QE is driven by the ZLB constraint itself or by the underlying asset-price dynamics during recessions. To address this, we solve a counterfactual version of the model that removes the ZLB constraint entirely ($R_t^n = R_{unc,t}^n$, allowing the nominal rate to fall below unity). We force the central bank to purchase assets for the same number of periods as in the ZLB baseline, so that the only difference is whether the nominal rate is constrained.

Table 8 reports the results. Without the ZLB, the Taylor rule can respond freely to the recession, allowing the nominal rate to fall below zero. This additional monetary accommodation moderates the recession—the output trough is slightly deeper in the no-ZLB counterfactual due to the absence of the rate floor interacting with the nonlinear dynamics, but the overall path of asset prices differs.

Table 8: Role of the Zero Lower Bound: PV of Central Bank Profits

Shock	QE Regime	Passive Unwind			Active Unwind		
		ZLB	No ZLB	Δ	ZLB	No ZLB	Δ
Risk Premium	Bond	-9.008	-7.448	1.560	-6.770	-5.032	1.738
	Equity	-5.034	-3.671	1.363	0.584	0.934	0.351
	Mixed	-7.021	-5.245	1.776	-2.905	-1.955	0.950

Notes: $\Delta = \text{No ZLB} - \text{ZLB Baseline}$. Positive values indicate that the ZLB worsens profitability. QE purchase periods in the no-ZLB model are forced to match the ZLB baseline duration.

The comparison reveals two findings. First, the profitability ranking is preserved without

the ZLB: equity QE continues to dominate bond QE across all configurations. This confirms that the fundamental mechanism—buying equities at depressed prices versus bonds at elevated prices—operates independently of the ZLB constraint.

Second, the ZLB amplifies the profitability differences. The column Δ (no-ZLB minus ZLB baseline) shows that removing the ZLB generally improves profitability, particularly for bond QE. This is because the ZLB compresses yields and inflates bond prices during the purchase phase, worsening the buy-high problem. Without the ZLB, bond prices are closer to their fundamental value during purchases, reducing (though not eliminating) the capital loss upon exit.

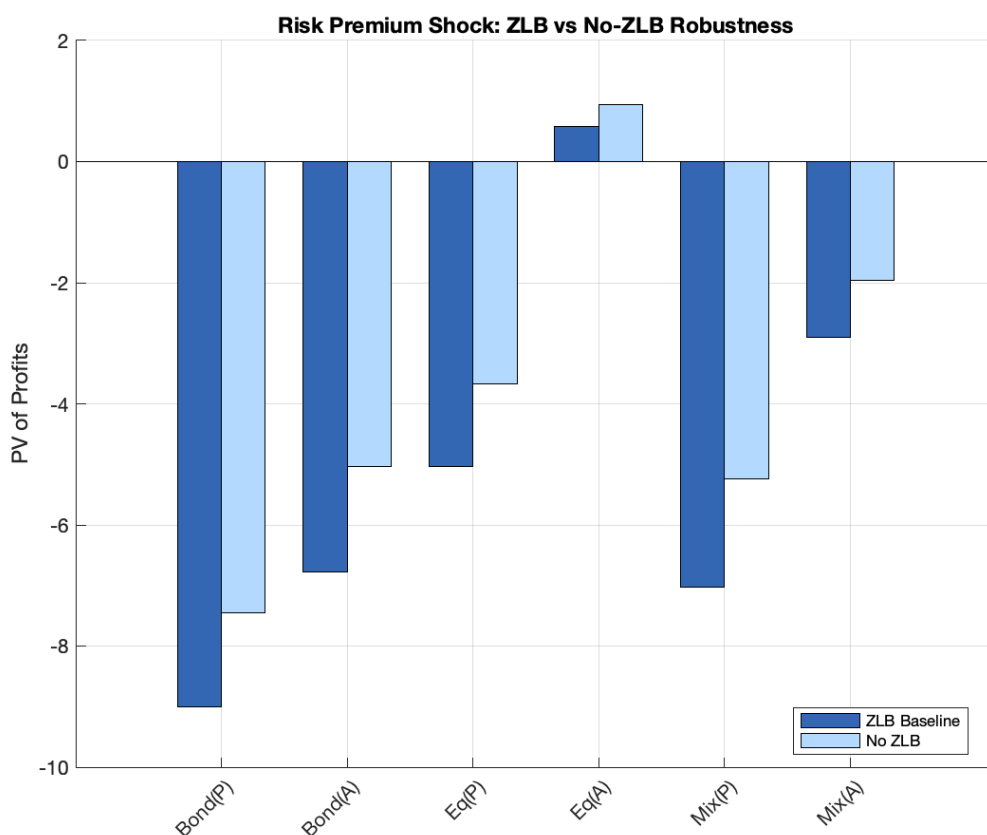


Figure 9: No-ZLB robustness check. PV of central bank profits with and without the ZLB constraint, for each QE regime and unwinding strategy under the risk premium shock. QE purchase periods are forced to match the ZLB baseline.

Figure 9 visualises the comparison. The pattern is clear: the profitability advantage of equity QE persists in both environments, but the absolute magnitude of losses is generally smaller without the ZLB. The implication is that the ZLB is a sufficient but not necessary condition for the equity QE profitability advantage—the result is driven by the cyclical

behaviour of asset prices during recessions, which the ZLB amplifies but does not create.

6 Discussion

6.1 Policy Implications

Our results suggest that central bank equity purchases can improve the profitability of QE programs, particularly when combined with active unwinding after the ZLB episode ends. Under demand-driven recessions—the primary scenario motivating QE—equity purchases exploit an asset-price asymmetry that bond purchases cannot: the central bank buys when equity prices are depressed and benefits from the recovery.

Should central banks buy equities? Our model provides a necessary but not sufficient case. Several considerations that the model abstracts from would need to be addressed before such a policy could be implemented.

Corporate governance. A central bank with significant equity holdings in listed firms would face governance questions. Should it exercise voting rights? How would it avoid conflicts of interest between its monetary policy mandate and its shareholder interests? The Bank of Japan addressed this by purchasing equity ETFs (index funds) rather than individual stocks, limiting direct corporate influence. Any equity purchase program would likely need to follow a similar passive, index-based approach.

Moral hazard. Central bank equity purchases could create moral hazard in equity markets if firms anticipate being “bailed out” through asset purchases during downturns. This concern is arguably smaller than it first appears: the central bank buys a broad index, not individual firms, and the purchase rule responds to macroeconomic conditions (the output gap), not firm-specific distress. Nevertheless, the potential for moral hazard—particularly if equity QE is anticipated—deserves attention in a richer model with heterogeneous firms.

Market distortion. Large-scale equity purchases could distort relative prices and resource allocation. If the central bank becomes a dominant shareholder in the equity market, it could reduce market discipline and price informativeness. The Bank of Japan’s experience is instructive: at its peak, the BoJ held roughly 7 percent of the Tokyo Stock Exchange market capitalisation through ETFs, raising concerns about market functioning and the “crowding out” of private price discovery.

Political economy. Equity purchases are more politically visible than bond purchases and could expose the central bank to political pressure. Gains on equity portfolios would be attributed to the central bank (and potentially create demands for distribution), while losses would invite criticism. The political economy of central bank equity holdings is substantially

more complex than that of government bond holdings, which are seen as a natural extension of monetary operations.

These considerations suggest that even if the profitability case for equity QE is strong in a structural model, practical implementation would require significant institutional design to mitigate governance, moral hazard, and political economy risks. A pragmatic middle ground might involve a mixed program (following the BoJ model) with a modest equity share, capturing some of the profitability advantage while limiting exposure to the concerns outlined above.

6.2 The Bank of Japan Experience

The Bank of Japan’s equity ETF purchase program provides the primary real-world precedent for central bank equity purchases. Beginning in 2010 and dramatically expanded under Quantitative and Qualitative Easing (QQE) from 2013, the BoJ accumulated approximately ¥37 trillion in equity ETF holdings by 2024. At its peak in 2020, the BoJ was purchasing ¥12 trillion per year in equity ETFs, making it one of the largest shareholders in the Japanese equity market.

Our model provides a structural explanation for the divergent portfolio outcomes of the BoJ (unrealised gains on equities) versus Western central banks (unrealised losses on bonds). The BoJ purchased equities during a prolonged period of low growth and depressed valuations. As Japanese equity markets recovered—particularly following the corporate governance reforms of the mid-2010s and the global equity rally—the BoJ’s portfolio appreciated substantially. By 2024, unrealised gains exceeded ¥20 trillion.

Meanwhile, the BoJ’s bond portfolio (like those of Western central banks) faced valuation pressures as global interest rates rose. The BoJ’s unique position of holding *both* equities and bonds meant that equity gains partially offset bond losses on its overall balance sheet—a diversification benefit that our model captures through the mixed QE regime.

Our model abstracts from several features of the Japanese experience that would matter for a complete calibration: the deflationary environment, the prolonged ZLB (decades rather than quarters), the role of yield curve control (which stabilised bond prices but created different distortions), and the equity market’s recovery dynamics. A Japan-specific calibration would be a valuable extension.

6.3 Federal Reserve Counterfactual

To complement the structural model analysis, we construct a simple partial-equilibrium counterfactual: what if the Federal Reserve had directed its QE purchase flows to the S&P 500

(via a broad equity index fund) instead of Treasury and agency MBS securities? We compute this counterfactual separately for each QE episode using actual SOMA holdings data from FRED and equity/bond total returns from ETF price data (SPY for equities; a 50/50 blend of TLT and IEF for bonds).⁴

For each episode, we identify monthly net purchases as positive changes in total SOMA holdings during the QE window and track each monthly tranche as a separate vintage. The bond portfolio is valued at current bond index prices (mark-to-market). The equity counterfactual considers three unwind assumptions: (i) *buy and hold*, where the central bank never sells; (ii) *match QT*, where equities are sold only during designated quantitative tightening windows at the same dollar pace as actual SOMA reductions; and (iii) *maturity-matched*, where equities are sold whenever SOMA declines (including passive bond maturity runoff between episodes).

Table 9: Federal Reserve Counterfactual: Bond vs. Equity QE Profit/Loss

Episode	Cost (\$B)	Bond P&L	Eq. (Hold)	Eq. (Match QT)	Eq. (Maturity)
QE1 (2008–10)	1,370	+678	+12,092	+12,092	+8,938
QE2 (2010–11)	637	+261	+3,754	+3,754	+1,125
QE3 (2012–14)	1,697	+267	+6,881	+5,709	+4,345
COVID (2020–22)	4,629	−1,047	+5,688	+4,527	+4,526
Total	8,332	+159	+28,416	+26,082	+18,934

Notes: All values in billions of US dollars. Bond P&L is mark-to-market on current bond index prices. Equity scenarios: Hold = never sell; Match QT = sell only during designated QT periods; Maturity = sell on any SOMA decline. Data through February 2026.

Table 9 reports the results. Across all four QE episodes, the Fed spent a total of \$8.3 trillion on bond purchases, generating a cumulative mark-to-market profit of just \$159 billion—dragged down by the COVID-era portfolio, which lost over \$1 trillion as rates rose sharply in 2022–2024. The equity counterfactual tells a dramatically different story. Even under the most conservative unwind assumption (maturity-matched selling), equities would have generated \$18.9 trillion in profit—over 100 times the bond outcome. The COVID episode is the most striking: bonds purchased at near-zero yields lost \$1.05 trillion, while equities bought during the March 2020 crash would have gained \$4.5 trillion.

Figure 10 shows the profit/loss comparison by episode and strategy. The pattern is consistent across all four episodes: equity returns dominate bond returns by an order of magnitude, with the largest gaps for QE1 and COVID—precisely the episodes where purchases occurred

⁴We use adjusted closing prices, which incorporate dividends and coupon payments. SOMA holdings are from the Federal Reserve’s H.4.1 release (FRED series TREAST and WSHOMCB).

during the deepest equity market troughs.

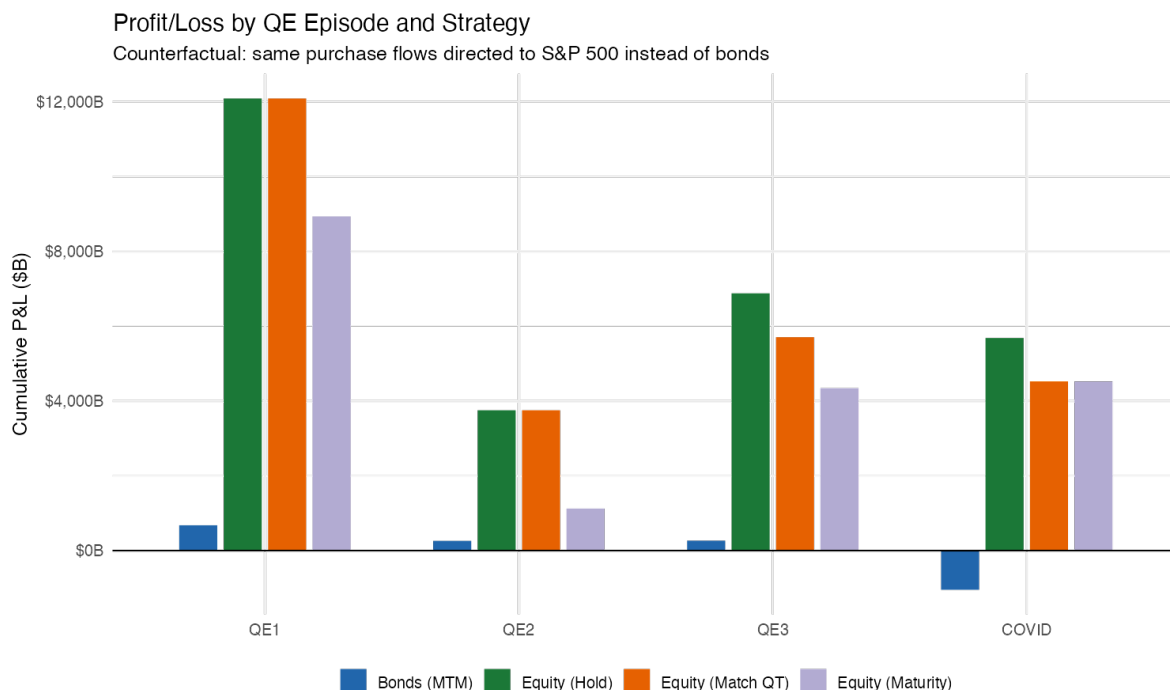


Figure 10: Profit/Loss by QE Episode and Strategy: Bond Actual vs. Equity Counterfactual

This exercise is deliberately simple and carries important caveats. It is a partial-equilibrium calculation that takes equity and bond prices as given; in general equilibrium, Fed equity purchases on this scale would substantially affect asset prices, potentially reducing the counterfactual gains. It also ignores the political and institutional constraints discussed in Section 6.1. Nevertheless, the exercise is informative because it illustrates the same buy-low-sell-high mechanism that drives our structural model results, applied to actual data. The fact that the profitability ranking—equities dominating bonds—holds across all four QE episodes, with different macroeconomic backdrops and equity market conditions, provides external validity for the model’s core prediction.

6.4 Limitations and Extensions

Several limitations of the analysis merit discussion.

Closed economy. The model is a closed economy, abstracting from exchange rate effects and international capital flows. Open economy considerations could either strengthen or weaken the case for equity QE: domestic equity purchases might attract capital inflows (appreciating the currency), while bond purchases might have the opposite effect through the yield curve. These general equilibrium effects could be important for small open economies.

Linear holding costs. Our model uses proportional holding costs (γ_S, γ_B) rather than quadratic portfolio adjustment costs. This simplification ensures a clean analytical steady state but limits the feedback from intermediary distress to household portfolio rebalancing. A consequence is that the financial shock has zero real effects in the current calibration—the linear holding cost does not amplify fire-sale dynamics. Quadratic costs or a richer intermediary-household interaction could generate larger effects for financial shocks and potentially activate the ZLB, making the profitability comparison relevant for a wider set of shocks.

Piecewise-linear approximation. The OccBin approach of [Guerrieri and Iacoviello \(2015\)](#) is a first-order approximation that handles the ZLB via piecewise-linear regime switching. This means that some shocks that might trigger the ZLB in a fully nonlinear model (e.g., monetary policy shocks with highly non-monotonic rate dynamics) do not reach the ZLB under the linearisation. Our results focus on the risk premium shock, which generates a clean ZLB episode under OccBin. The qualitative profitability ranking—equity QE dominating bond QE—is driven by the asset-price asymmetry (buying equities at depressed prices versus bonds at elevated prices) rather than by nonlinearities, and is therefore robust to the solution method. The active unwind is implemented as a partial-equilibrium adjustment to the passive OccBin solution, which is valid given the modest macroeconomic scale of QE in the model.

QE scale and fiscal feedback. The calibrated purchase intensities generate central bank balance sheets of approximately 270 percent of quarterly GDP (68 percent of annual GDP) at peak—comparable to the Federal Reserve’s peak balance sheet of roughly 36 percent of annual GDP. The distortionary labor tax creates a notable feedback channel: QE losses raise taxes, which reduce output, deepening the trough from -3.4 percent (no QE) to -5.0 percent (bond QE). While this fiscal amplification is a genuine general-equilibrium effect, its magnitude depends on the tax rule parameters (ρ_τ, ϕ_τ) and the remittance passthrough (ϕ_{Rem}) .

TANK versus HANK. Our two-agent structure captures the key role of market segmentation and the fiscal channel but abstracts from the richer distributional dynamics of a heterogeneous-agent New Keynesian (HANK) model. In a HANK framework with a continuous wealth distribution, the effects of QE composition on portfolio returns would propagate through the wealth distribution in complex ways. [Sims et al. \(2024\)](#) show that HANK amplifies some QE transmission channels while dampening others; incorporating equity composition into a HANK-QE framework is a natural extension.

Time-varying risk premia. The model generates endogenous but relatively stable credit spreads through the intermediary leverage constraint. In reality, risk premia are highly time-

varying, particularly during crises. A model with stochastic volatility or disaster risk could generate richer dynamics in which the central bank’s timing of equity purchases—buying during crisis-level risk premia—produces even larger profitability gains.

Estimation scope. We estimate a subset of the model’s parameters (monetary policy rule, risk aversion, shock processes) using Bayesian methods on US macroeconomic data (Section 3.4). However, several parameters central to the QE mechanism—the market segmentation parameter ψ_u , the holding cost parameters γ_S and γ_B , the portfolio balance elasticities κ_S and κ_B , and the QE rule coefficients—are calibrated rather than estimated, as they have limited identification from pre-QE aggregate data. Estimating these parameters using data from QE episodes across multiple countries, or using micro-level data on portfolio holdings, would further discipline the quantitative results.

7 Conclusion

This paper asks whether the composition of central bank asset purchases—long-term government bonds versus equities—systematically affects central bank profitability and, through the fiscal channel, household welfare. We study this question in a New Keynesian DSGE model with two household types, Gertler-Karadi financial intermediaries with asset-specific divertability, and an explicit central bank balance sheet with profit accounting.

The answer is yes, and the mechanism is an asset-price asymmetry at the zero lower bound. Bond prices are elevated during ZLB episodes (because yields are compressed to near-zero levels), so the central bank buys high and subsequently sells low—or, under passive unwinding, earns below-market returns on a shrinking portfolio. Equity prices are depressed (because investment prospects deteriorate during recessions), so the central bank buys low and benefits from capital appreciation during the recovery.

Under a risk premium shock generating a 6-quarter ZLB episode, equity QE with passive unwinding reduces present-value losses by 44 percent relative to bond QE. With active unwinding, equity QE generates a *net profit*, while bond QE still produces substantial losses. QE composition also matters for the real economy: the fiscal cost of QE losses feeds back through higher distortionary taxes, deepening the recession by up to 1.6 percentage points under bond QE relative to no QE, but only 0.9 percentage points under equity QE. Welfare effects systematically favour restricted (hand-to-mouth) households under equity QE, operating through the fiscal channel: more profitable QE means lower taxes, which disproportionately benefits those who cannot smooth the tax burden through financial markets. A counterfactual exercise using actual Federal Reserve SOMA purchase data confirms that the buy-low-sell-high logic holds empirically: equity returns would have dominated bond returns

by an order of magnitude across all four QE episodes.

These results provide a structural explanation for the divergent portfolio outcomes of the Bank of Japan's equity purchases (large unrealised gains) versus Western central banks' bond-heavy QE programs (large unrealised losses). The asset-price asymmetry is a feature of the macroeconomic environment at the ZLB, not an accident of market timing.

For central bankers contemplating the next recession that drives interest rates to their effective lower bound, our findings suggest that the composition of unconventional policy matters—not just for macroeconomic stabilisation, but for the central bank's own financial position and, ultimately, for the fiscal burden borne by taxpayers. The practical challenges of equity purchases are real, but the profitability case deserves serious consideration alongside the governance and political economy concerns that have traditionally kept equities off central bank balance sheets.

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Appendix A Full Model Derivations

A.1 Unrestricted Household FOCs

The unrestricted household maximises

$$\max_{\{C_{u,t}, L_{u,t}, D_{u,t}, S_{u,t}, B_{u,t}^L\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[A_{pref,t} \frac{C_{u,t}^{1-\sigma_c}}{1-\sigma_c} - \zeta_U \frac{L_{u,t}^{1+\varphi}}{1+\varphi} \right]$$

subject to the budget constraint (2). The preference shock $A_{pref,t}$ enters only the consumption term (separable utility). Deposits $D_{u,t}$ are a choice variable in the Lagrangian but are eliminated via Walras' Law in the implemented model.

Define the Lagrangian with multiplier $\Lambda_{u,t}$ on the period- t budget constraint:

$$\begin{aligned} \mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t & \left\{ A_{pref,t} \frac{C_{u,t}^{1-\sigma_c}}{1-\sigma_c} - \zeta_U \frac{L_{u,t}^{1+\varphi}}{1+\varphi} \right. \\ & + \Lambda_{u,t} \left[W_t L_{u,t} + (Z_{k,t} + (1-\delta)Q_t^K) S_{u,t-1} + \frac{1+\chi Q_t^L}{\Pi_t} B_{u,t-1}^L + R_{t-1} D_{u,t-1} \right. \\ & \left. \left. - C_{u,t} - Q_t^K (1+\gamma_S) S_{u,t} - Q_t^L (1+\gamma_B) B_{u,t}^L - D_{u,t} - T_t + \text{profits}_t \right] \right\} \end{aligned}$$

The first-order conditions are:

Consumption ($\partial \mathcal{L} / \partial C_{u,t} = 0$):

$$A_{pref,t} \cdot C_{u,t}^{-\sigma_c} = \Lambda_{u,t} \equiv UC_{u,t} \quad (\text{A.1})$$

Labour ($\partial \mathcal{L} / \partial L_{u,t} = 0$): Since $A_{pref,t}$ does not multiply the labour disutility:

$$W_t \cdot UC_{u,t} = \zeta_U \cdot L_{u,t}^\varphi \quad (\text{A.2})$$

Deposits ($\partial \mathcal{L} / \partial D_{u,t} = 0$):

$$UC_{u,t} = \beta \cdot R_t \cdot \mathbb{E}_t[UC_{u,t+1}] \quad (\text{A.3})$$

Equity ($\partial \mathcal{L} / \partial S_{u,t} = 0$):

$$UC_{u,t} \cdot Q_t^K \left(1 + \gamma_S \exp \left(-\kappa_S \frac{S_{t-1}^{CB}}{K_{t-1}} \right) \right) v_{rp,t} = \beta \cdot \mathbb{E}_t [UC_{u,t+1} (Z_{k,t+1} + (1-\delta)Q_{t+1}^K)] \quad (\text{A.4})$$

The portfolio balance term $\gamma_S \exp(-\kappa_S S_{t-1}^{CB}/K_{t-1})$ captures the reduction in household hold-

ing costs when the central bank absorbs a larger share of equities. The risk premium shifter $v_{rp,t}$ follows an AR(1) process.

Long bonds ($\partial \mathcal{L} / \partial B_{u,t}^L = 0$):

$$UC_{u,t} \cdot Q_t^L \left(1 + \gamma_B \exp \left(-\kappa_B \frac{B_{t-1}^{CB}}{B_{t-1}^L} \right) \right) = \beta \cdot \mathbb{E}_t \left[UC_{u,t+1} \cdot \frac{1 + \chi Q_{t+1}^L}{\Pi_{t+1}} \right] \quad (\text{A.5})$$

The stochastic discount factor is defined as $\Lambda_{u,t} \equiv \beta \cdot UC_{u,t+1} / UC_{u,t}$.

Steady-state implications. From (A.3): $R_{ss} = 1/\beta$. At steady state, $S_{ss}^{CB} = B_{ss}^{CB} = 0$, so the portfolio balance terms equal $\exp(0) = 1$. From (A.4): $R_{ss}^{eq} = (1 + \gamma_S)/\beta$, yielding the equity spread γ_S/β . From (A.5): $R_{ss}^L = (1 + \gamma_B)/\beta$, yielding the bond spread γ_B/β . The holding cost calibration $\gamma_B = \gamma_S \cdot \lambda_L / \lambda_{eq}$ ensures $\text{spread}^{eq} / \text{spread}^L = \lambda_{eq} / \lambda_L$. The portfolio balance parameters κ_S and κ_B govern the sensitivity of holding costs to central bank market share; larger values amplify the spread compression from QE purchases.

A.2 Restricted Household FOCs

Restricted households maximise the same objective (9) subject to the hand-to-mouth budget constraint (10). Since there is no intertemporal choice, the Lagrangian reduces to a static problem each period.

Consumption:

$$UC_{r,t} = A_{pref,t} \cdot C_{r,t}^{-\sigma_c} \quad (\text{A.6})$$

Labour:

$$W_t \cdot UC_{r,t} = \zeta_R \cdot L_{r,t}^\varphi \quad (\text{A.7})$$

Budget:

$$C_{r,t} = W_t L_{r,t} + \text{profits}_t - T_t \quad (\text{A.8})$$

The labour disutility weight ζ_R is calibrated so that $L_{r,ss} = 1$: $\zeta_R = W_{ss} \cdot C_{r,ss}^{-\sigma_c}$.

A.3 Rotemberg Pricing and the NKPC

Intermediate firm i maximises the present value of profits subject to the demand curve $Y_{i,t} = (P_{i,t}/P_t)^{-\epsilon} Y_t$ and Rotemberg quadratic price adjustment costs $(\psi_p/2)(\Pi_{i,t} - 1)^2 Y_t$. Including the production subsidy $(1 + \tau_S)$, the firm's problem in symmetric equilibrium (with $P_{i,t} = P_t$) reduces to choosing inflation Π_t to maximise:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \Lambda_{0,t} \left[(1 + \tau_S) Y_t - MC_t Y_t - \frac{\psi_p}{2} (\Pi_t - 1)^2 Y_t \right]$$

The first-order condition with respect to Π_t yields the NKPC:

$$\psi_p \Pi_t (\Pi_t - 1) = (1 + \tau_S)(1 - \epsilon) + \epsilon \cdot MC_t + \beta \frac{UC_{u,t+1}}{UC_{u,t}} \psi_p \Pi_{t+1} (\Pi_{t+1} - 1) \frac{Y_{t+1}}{Y_t} \quad (\text{A.9})$$

The Rotemberg parameter ψ_p is linked to the Calvo probability θ by $\psi_p = \theta(\epsilon - 1)/[(1 - \theta)(1 - \beta\theta)]$. With $\theta = 0.75$, $\epsilon = 11$, $\beta = 0.99$: $\psi_p \approx 116.5$.

A.4 Capital Goods Producer and Tobin's Q

The capital goods producer maximises the present value of profits from selling new capital at price Q_t^K net of investment costs:

$$\max_{\{I_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \Lambda_{0,t} \left[Q_t^K \left(1 - \frac{\zeta_{inv}}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \right) I_t - I_t \right]$$

Let $x_t \equiv I_t/I_{t-1}$. The FOC with respect to I_t , noting that both x_t and $x_{t+1} = I_{t+1}/I_t$ depend on I_t :

$$Q_t^K \left[1 - \frac{\zeta_{inv}}{2} (x_t - 1)^2 - \zeta_{inv} (x_t - 1) x_t \right] + \mathbb{E}_t [\Lambda_{u,t} Q_{t+1}^K \zeta_{inv} (x_{t+1} - 1) x_{t+1}^2] = 1 \quad (\text{A.10})$$

At steady state ($x = 1$): $Q_{ss}^K = 1$.

Capital accumulation:

$$K_t = \left[1 - \frac{\zeta_{inv}}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \right] I_t + (1 - \delta) K_{t-1} \quad (\text{A.11})$$

A.5 Financial Intermediary Problem

A banker maximises the expected present value of terminal net worth:

$$V_t = \mathbb{E}_t \sum_{j=0}^{\infty} (1 - \sigma_b) \sigma_b^j \cdot \beta^{j+1} \cdot \Lambda_{u,t+1+j} \cdot N_{fi,t+1+j}$$

subject to the incentive constraint $V_t \geq \lambda_{eq} Q_t^K S_{fi,t} + \lambda_L Q_t^L B_{fi,t}^L$.

Value function conjecture: $V_t = \nu_t^S Q_t^K S_{fi,t} + \nu_t^B Q_t^L B_{fi,t}^L + \eta_t N_{fi,t}$.

Bellman equation:

$$V_t = \mathbb{E}_t [\beta \Lambda_{u,t+1} ((1 - \sigma_b) N_{fi,t+1} + \sigma_b V_{t+1})]$$

Using the net worth evolution and $D_t = A_t - N_{fi,t}$ (where $A_t = Q_t^K S_{fi,t} + Q_t^L B_{fi,t}^L$):

$$N_{fi,t+1} = (R_{t+1}^{eq} - R_t)Q_t^K S_{fi,t} + (R_{t+1}^L - R_t)Q_t^L B_{fi,t}^L + R_t N_{fi,t}$$

Matching coefficients yields:

$$\nu_t^S = \mathbb{E}_t[\beta \Lambda_{u,t+1} \Omega_{fi,t+1}](R_{t+1}^{eq} - R_t) \quad (\text{A.12})$$

$$\nu_t^B = \mathbb{E}_t[\beta \Lambda_{u,t+1} \Omega_{fi,t+1}](R_{t+1}^L - R_t) \quad (\text{A.13})$$

$$\eta_t = \mathbb{E}_t[\beta \Lambda_{u,t+1} \Omega_{fi,t+1}] R_t \quad (\text{A.14})$$

where $\Omega_{fi,t+1} = (1 - \sigma_b) + \sigma_b(\nu_{t+1}^S \phi_{fi,t+1} x_{fi} + \nu_{t+1}^B \phi_{fi,t+1} (1 - x_{fi}) + \eta_{t+1})$.

Binding incentive constraint. With fixed portfolio composition $x_{fi} = x_{fi,ss}$, define $\bar{\lambda} = \lambda_{eq} x_{fi} + \lambda_L (1 - x_{fi})$. The binding IC yields:

$$\phi_{fi,t} = \frac{\eta_t}{\bar{\lambda} - \nu_t^S x_{fi} - \nu_t^B (1 - x_{fi})} \quad (\text{A.15})$$

The Omega recursion implemented in the model is:

$$\Omega_{fi,t} = (1 - \sigma_b) + \sigma_b \Lambda_{u,t+1} \Omega_{fi,t+1} [\text{sp}_t^{eq} \phi_{fi,t} x_{fi} + \text{sp}_t^L \phi_{fi,t} (1 - x_{fi}) + R_t] \quad (\text{A.16})$$

Steady state. The nonlinear steady-state system reduces to a single equation in $\theta_\mu \equiv \mu/(1 + \mu)$:

$$f(\theta_\mu) = \frac{(1 - \sigma_b)(1 - \theta_\mu)}{1 - \theta_\mu - \sigma_b} - \frac{\lambda_{eq} \theta_\mu}{\gamma_S} = 0 \quad (\text{A.17})$$

with root in $(0, 1 - \sigma_b)$, solved numerically via bisection.

A.6 CB Profit Accounting

The central bank's operating profit decomposes as:

$$\begin{aligned} \text{ProfCB}_t = & \underbrace{R_t^{eq} Q_{t-1}^K S_{t-1}^{CB} + R_t^L Q_{t-1}^L B_{t-1}^{CB}}_{\text{Gross asset returns}} - \underbrace{R_{t-1} \text{Res}_{t-1}}_{\text{Reserve interest}} \\ & - \underbrace{Q_t^K (S_t^{CB} - S_{t-1}^{CB}) + Q_t^L (B_t^{CB} - B_{t-1}^{CB})}_{\text{Net new purchases}} \end{aligned} \quad (\text{A.18})$$

This can be further decomposed into:

Net interest income:

$$\text{NII}_t = Z_{k,t} S_{t-1}^{CB} + \frac{B_{t-1}^{CB}}{\Pi_t} - R_{t-1} \text{Res}_{t-1}$$

Capital gains:

$$\text{CapGain}_t = (Q_t^K - Q_{t-1}^K) S_{t-1}^{CB} + (\chi Q_t^L / \Pi_t - Q_{t-1}^L) B_{t-1}^{CB}$$

The present value of profits is $PV^{CB} = \sum_{j=0}^T R_{ss}^{-j} \text{ProfCB}_{t+j}$.

A.7 Complete Steady-State Solution

The steady-state computation proceeds in six steps:

Step 1: Prices and rates.

$$\begin{aligned} \Pi_{ss} &= 1.005, \quad R_{ss} = 1/\beta, \quad R_{ss}^n = \Pi_{ss}/\beta \\ R_{ss}^{eq} &= (1 + \gamma_S)/\beta, \quad R_{ss}^L = (1 + \gamma_B)/\beta \\ Q_{ss}^K &= 1, \quad Z_{k,ss} = R_{ss}^{eq} - 1 + \delta \\ R_{ss}^{Ln} &= R_{ss}^L \cdot \Pi_{ss}, \quad Q_{ss}^L = 1/(R_{ss}^{Ln} - \chi) \\ MC_{ss} &= [(1 + \tau_S)(\epsilon - 1) + \psi_p \Pi_{ss}(\Pi_{ss} - 1)(1 - \beta)]/\epsilon \end{aligned}$$

Step 2: Financial intermediary. Solve $f(\theta_\mu) = 0$ (equation A.17) for θ_μ . Then:

$$\eta_{ss} = \frac{\lambda_{eq} \theta_\mu}{\gamma_S}, \quad \Omega_{fi,ss} = \frac{1 - \sigma_b}{1 - \sigma_b \beta \bar{R}_\phi}$$

Step 3: Output and capital.

$$K/Y = \alpha \cdot MC_{ss}/Z_{k,ss}, \quad Y_{ss} = (K/Y)^{\alpha/(1-\alpha)}, \quad K_{ss} = (K/Y) \cdot Y_{ss}, \quad I_{ss} = \delta K_{ss}$$

Step 4: Portfolio allocation. With $s_u^{eq} = s_u^{bond} = 0.30$:

$$\begin{aligned} S_{fi,ss} &= (1 - s_u^{eq}) K_{ss}, \quad B_{fi,ss}^L = (1 - s_u^{bond}) B_{ss}^L \\ x_{fi,ss} &= S_{fi,ss} / (S_{fi,ss} + Q_{ss}^L B_{fi,ss}^L), \quad \phi_{fi,ss} = \eta_{ss} / [(1 - \theta_\mu) \bar{\lambda}] \end{aligned}$$

Step 5: Aggregates.

$$C_{ss} = Y_{ss} - I_{ss} - G_{ss} - (\psi_p/2)(\Pi_{ss} - 1)^2 Y_{ss}$$

Step 6: Household consumption and labour weights.

$$C_{r,ss} = W_{ss} + \text{profits}_{ss} - T_{ss}, \quad C_{u,ss} = (C_{ss} - (1 - \psi_u)C_{r,ss})/\psi_u$$

$$\zeta_U = W_{ss} \cdot C_{u,ss}^{-\sigma_c}, \quad \zeta_R = W_{ss} \cdot C_{r,ss}^{-\sigma_c}$$

Appendix B Computational Details

Software. The model is solved using Dynare 6.1 within MATLAB R2024a on Apple Silicon macOS.

Solution method. The baseline experiments use the OccBin piecewise-linear solver (Guerrieri and Iacoviello, 2015) implemented in Dynare 6.1, which handles the ZLB as an occasionally binding constraint via regime switching. QE purchases are activated via exogenous indicator variables when the ZLB binds. Active unwinding is implemented as a post-processing adjustment to the passive OccBin solution: after solving for equilibrium paths with passive unwinding, we recompute central bank balance sheet variables (B_t^{CB} , S_t^{CB} , ProfCB_t) using the equilibrium prices from the passive solution. This partial-equilibrium approach is valid given the modest macroeconomic scale of QE in the model.

Convergence. The OccBin solver uses a 200-period simulation horizon with piecewise-linear regime switching. All 56 experiments (4 QE regimes $\times$ 7 shocks $\times$ 2 unwind strategies) converge.

Welfare computation. Consumption-equivalent variation (CEV) is computed by comparing lifetime utility under QE to lifetime utility under no-QE for the same shock. Let U_j^{QE} and U_j^{noQE} denote the discounted lifetime utilities of household type $j \in \{u, r\}$. The CEV Δ_j solves:

$$U_j^{noQE}(1 + \Delta_j) = U_j^{QE}$$

where the scaling applies to the consumption stream. For CRRA utility, $\Delta_j = \exp[(U_j^{QE} - U_j^{noQE})(1 - \sigma_c)/U_j^{noQE,C}] - 1$, with $U_j^{noQE,C}$ the consumption component of utility.

Profit calculations. Present-value profits are computed as $PV^{CB} = \sum_{t=0}^{200} (1/R_{ss})^t \cdot \text{ProfCB}_t$. Peak loss is $\min_t \sum_{s=0}^t \text{ProfCB}_s$. Total remittances is $\sum_{t=0}^{200} \text{Rem}_t$.

Appendix C Model Equations Summary

The model comprises 49 endogenous variables and 49 equations. The core 43 equations of the stochastic model are:

- (1) Marginal utility, unrestricted: $UC_u = A_{pref} C_u^{-\sigma_c}$

- (2) Labour supply, unrestricted: $W \cdot UC_u = \zeta_U L_u^\varphi$
- (3) Euler (deposits), unrestricted: $UC_u = \beta R \cdot UC_u(+1)$
- (4) Euler (equity) with portfolio balance: $UC_u Q^K (1 + \gamma_S e^{-\kappa_S S_{-1}^{CB}/K-1}) v_{rp} = \beta UC_u(+1) [Z_k(+1) + (1 - \delta) Q^K(+1)]$
- (5) Euler (long bond) with portfolio balance: $UC_u Q^L (1 + \gamma_B e^{-\kappa_B B_{-1}^{CB}/B_{-1}^L}) = \beta UC_u(+1) (1 + \chi Q^L(+1)) / \Pi(+1)$
- (6) SDF: $\Lambda_u = \beta UC_u(+1) / UC_u$
- (7) Marginal utility, restricted: $UC_r = A_{pref} C_r^{-\sigma_c}$
- (8) Labour supply, restricted: $W \cdot UC_r = \zeta_R L_r^\varphi$
- (9) Budget, restricted: $C_r = W(1 - \tau^L) L_r + \text{profits}$
- (10) Production function: $Y = Z_{tfp} K(-1)^\alpha L_{agg}^{1-\alpha}$
- (11) Capital accumulation with investment efficiency: $K = \Upsilon_{inv} [1 - \zeta_{inv} / 2 (I/I(-1) - 1)^2] I + (1 - \delta) K(-1)$
- (12) Tobin's Q with investment efficiency
- (13) Capital demand: $Z_k = \alpha \cdot MC \cdot Y / K(-1)$
- (14) Labour demand: $W = (1 - \alpha) MC \cdot Y / L_{agg}$
- (15) NKPC (Rotemberg)
- (16) Profits: $\text{profits} = (1 - MC - \psi_p (\Pi - 1)^2 / 2) Y$
- (17) Equity return: $R^{eq} = (Z_k + (1 - \delta) Q^K) / Q^K(-1)$
- (18) Nominal bond return: $R^{Ln} = (1 + \chi Q^L) / Q^L(-1)$
- (19) Real bond return: $R^L = R^{Ln} / \Pi$
- (20) Fisher equation: $R = R^n / \Pi(+1)$
- (21) Equity spread: $\text{sp}^{eq} = R^{eq}(+1) - R$
- (22) Bond spread: $\text{sp}^L = R^L(+1) - R$
- (23) Taylor rule
- (24) FI net worth evolution
- (25) FI balance sheet: $D = (\phi_{fi} - 1) N_{fi}$
- (26) FI leverage: $\phi_{fi} N_{fi} = Q^K S_{fi} + Q^L B_{fi}^L$
- (27) FI equity share definition
- (28) FI fixed composition: $x_{fi} = x_{fi,ss}$
- (29) Omega recursion
- (30) Incentive constraint (binding)
- (31) Tax rule: $\tau^L = \rho_\tau \tau^L(-1) + (1 - \rho_\tau) [\tau_{ss}^L + \phi_\tau (Q^L B^L / Y - Q_{ss}^L B_{ss}^L / Y_{ss}) - \text{Rem} / (W \cdot L_{agg})]$
- (32) Government spending: AR(1) with shock ε^G
- (33) Bond supply: AR(1)
- (34) Aggregate consumption

- (35) Aggregate labour
- (36) Resource constraint
- (37) Equity market clearing
- (38) Bond market clearing
- (39) –43. Shock processes (Z_{tfp} , A_{pref} , ξ_{fin} , v_{rp} , Υ_{inv})

The OccBin model adds 6 variables for the CB balance sheet (B^{CB} , S^{CB} , ProfCB, Rem, Div, and an unconstrained rate $R^{n,unc}$), yielding 49 equations total. The ZLB constraint is $R_t^n = \max(1, R_t^{n,unc})$.

Appendix D Bayesian Estimation Diagnostics

This section presents diagnostic information for the Bayesian estimation described in Section 3.4.

D.1 Prior and Posterior Distributions

Figures 11 and 12 display the prior (grey) and posterior (black) marginal densities for all estimated parameters. The posterior distributions are computed from the Metropolis–Hastings draws (100,000 per chain, 2 chains). Several features are notable. First, the data are informative for most parameters: the posterior is substantially tighter than the prior for ϕ_π , ρ_r , σ_c , and all shock standard deviations. Second, ρ_A (preference shock persistence) piles up near its upper bound of 0.999, consistent with the near-unit-root behaviour discussed in the text. Third, the mode (green vertical line) and posterior mean (dashed) are closely aligned for all parameters, indicating symmetric posteriors.

D.2 MCMC Convergence Diagnostics

Figure 13 shows the multivariate MCMC diagnostic, which tracks convergence of the within-chain and between-chain moments. The diagnostic shows convergence by approximately 30,000 draws. The overall acceptance rate is 17% (within the standard 15–35% range for random-walk Metropolis–Hastings), and inefficiency factors range from 50 to 200.

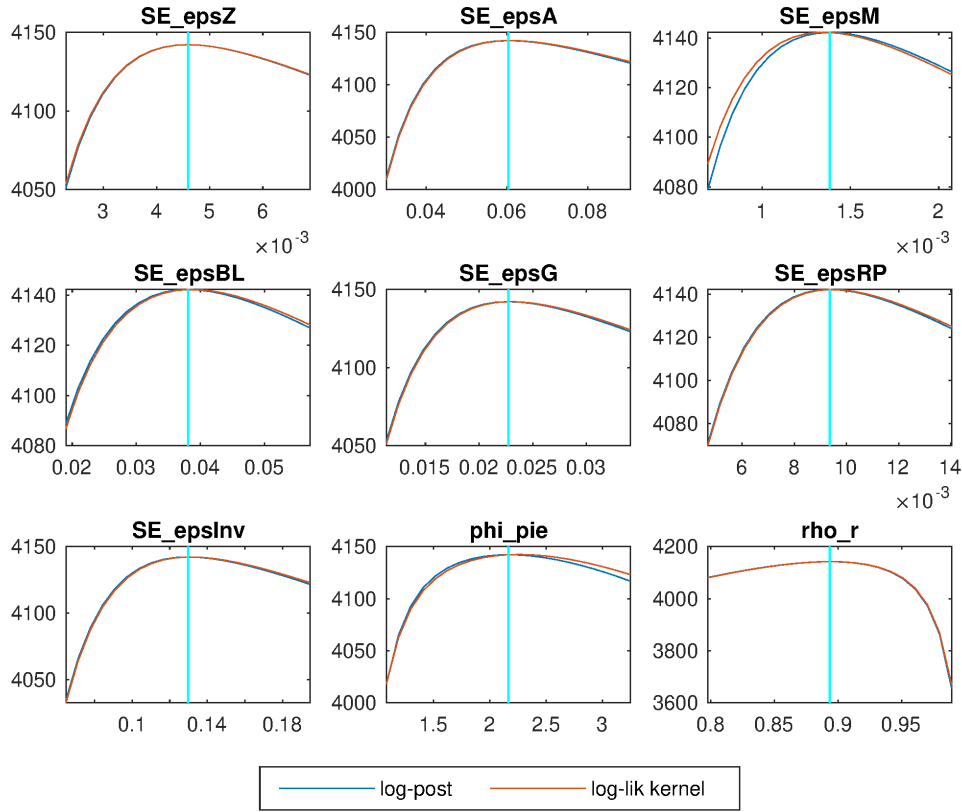


Figure 11: Prior and posterior marginal densities (page 1 of 2). Grey: prior density. Black: posterior density from MCMC. Green vertical line: posterior mode.

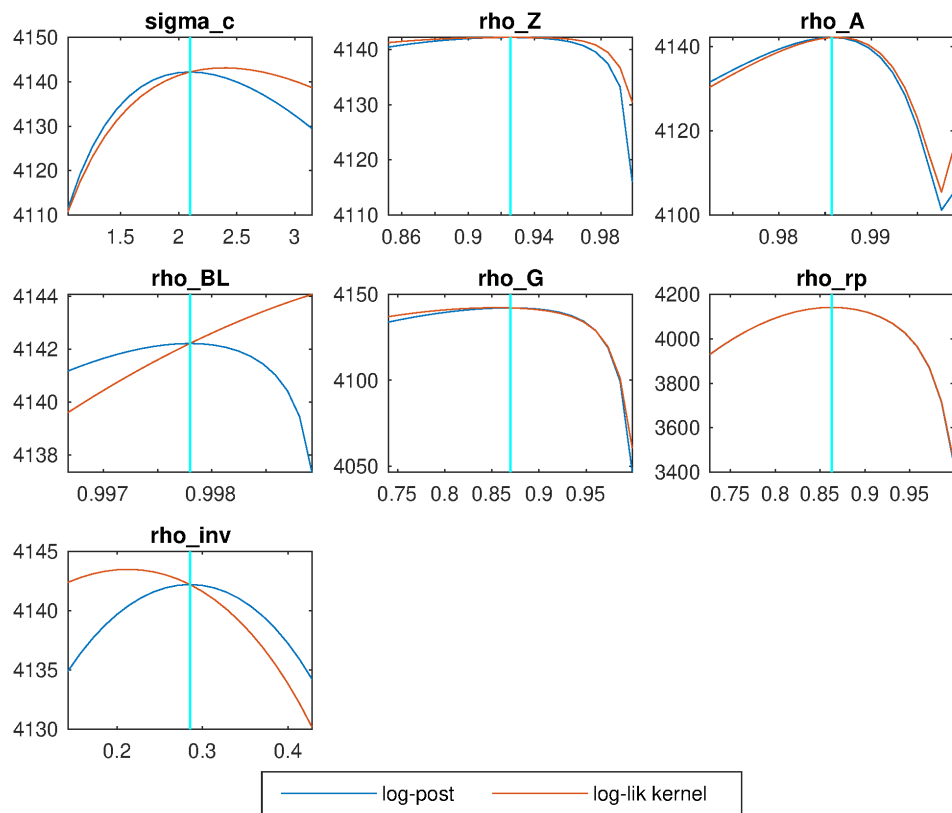


Figure 12: Prior and posterior marginal densities (page 2 of 2).

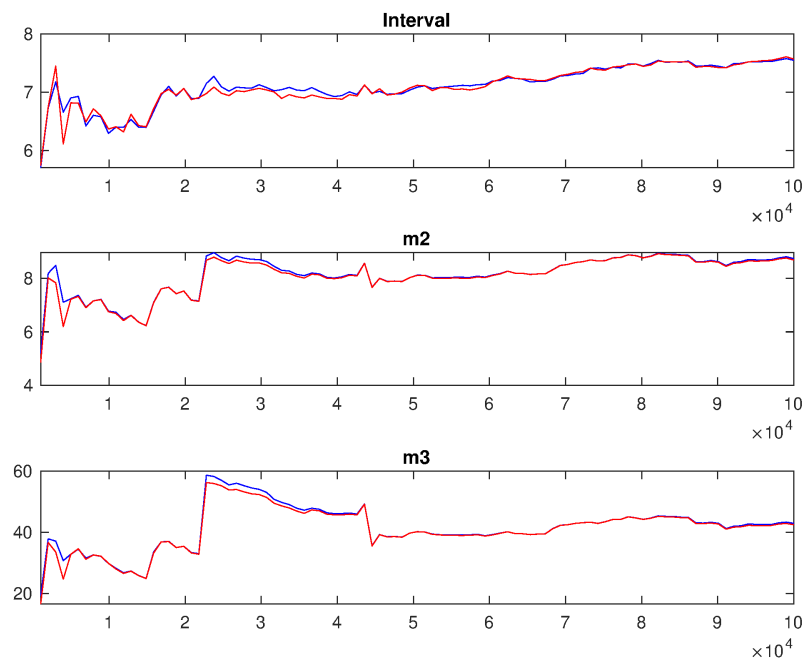


Figure 13: Multivariate MCMC convergence diagnostic. Blue: within-chain moment. Red: between-chain moment. Convergence is achieved when both lines stabilise.